Portfolio

Data Analyst

Google Certified

Maria Aslam

Hi, I am Maria Aslam A Data Analyst & BI Developer

I am Data Analyst & BI Developer focused on turning complex data into meaningful insights that drive informed decision-making.

About Me

Personal Info

First name: Maria

Last name: Aslam

Nationality: Pakistani

City: Lahore

Certificates

- Professional Data Analytics

  By Google

- Data Analytics & Business Intelligence

  By DigiSkills

Education

- Bachelor’s in Physics | 2021 - 2025

  Lahore College for Women University, Lahore

Skills & Tool

Data & Analytics Tools

- Power BI
- Tableau
- Excel
- Google Sheets,

Programming & Scripting

- R (tidyverse, dplyr, ggplot2, readr)
- Python (Pandas, NumPy, Matplotlib
  Seaborn, Plotly)

Database & Querying

- SQL (BigQuery)

Experience

- ## Freelance Data Analyst | Dec 2025 Present

  Worked on end-to-end data analysis projects. Performed data cleaning, analysis, & dashboard creation. Delivered insights to support data-driven decisions and performance improvement

- ## Teaching Assistant & Data Support | Part time | Jan 2023 - Jan 2024

  Maintained & organized financial and administrative records using MS Excel. Handled and structured data to improve accuracy & reporting efficiency. Supported data tracking and record management in an academic environment.

PROJECTS

Executive Supply Chain Performance | Q1 2025

Total Revenue
$599,338

Total Net Profit
$312,982

Global Defect Rate
! 2.29%

Avg. Profit Margin
54%

Units Sold
47272

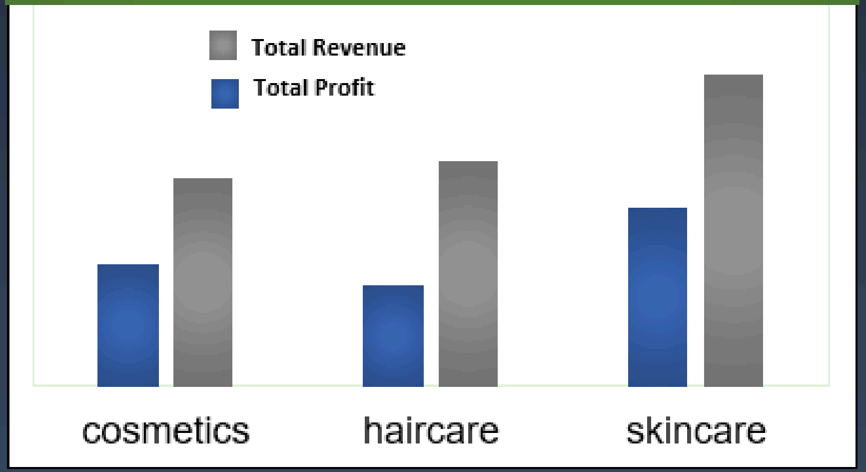
Slicers

Select Region
- East Gateway Hub
- Northern Logistics Node
- South Asia Mfg Hub
- Tech & R&D Center
- West Coast Hub (Port)

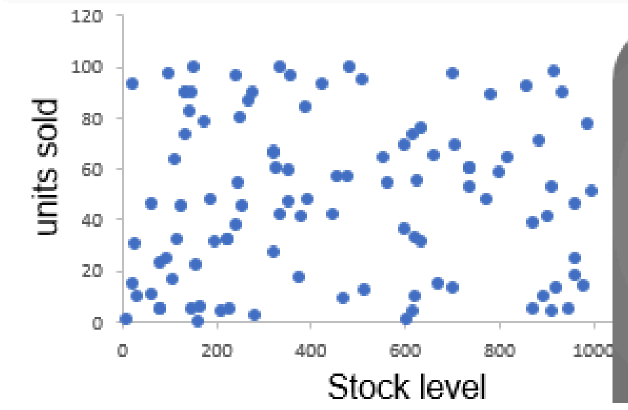
Select Product type
- cosmetics
- haircare
- skincare

Product Performance

Revenue & Profit by Product type



Units Sold Vs. Stock Level



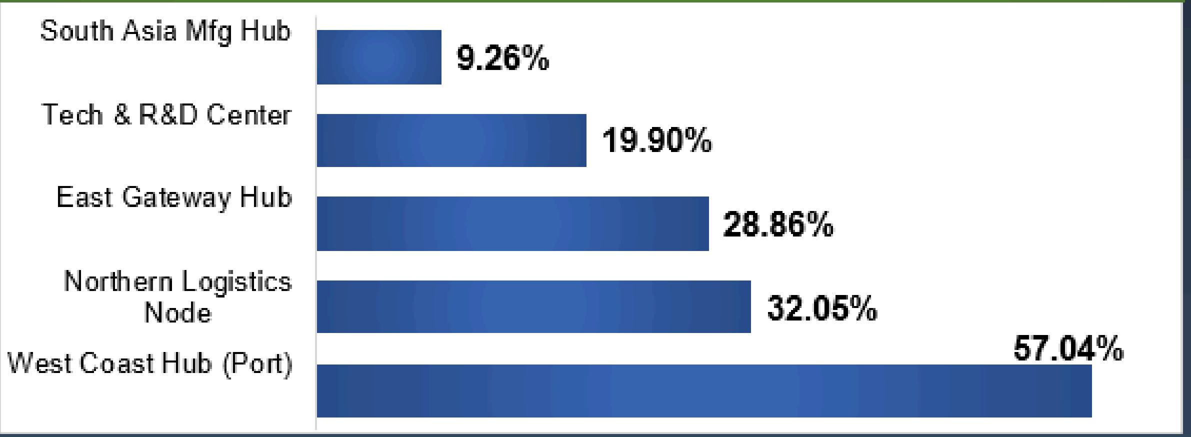
Stockout Alert: High-demand SKUs (top-left) require immediate restock.

Overstock Risk: Low-velocity "Zombie" items (bottom-right) are wasting warehouse space.

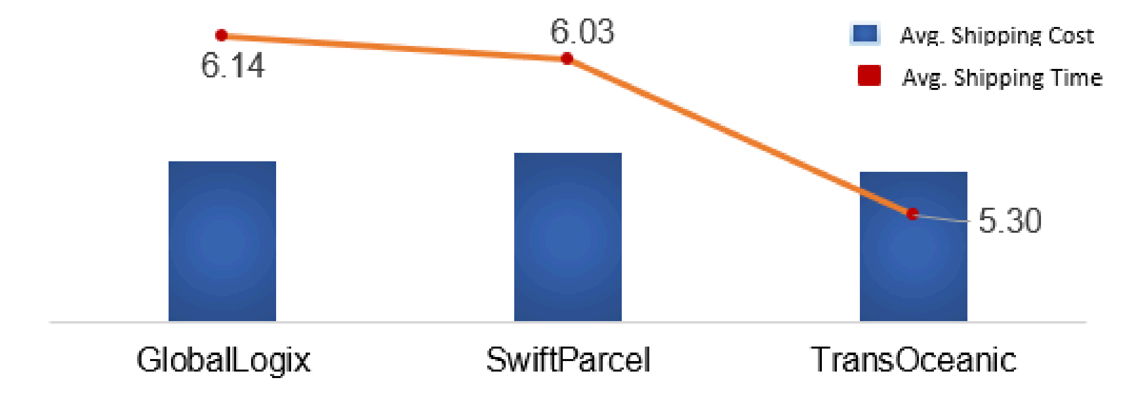
Action: Liquidate bottom-right stock to free up capital.

Operational Performance & Quality Control

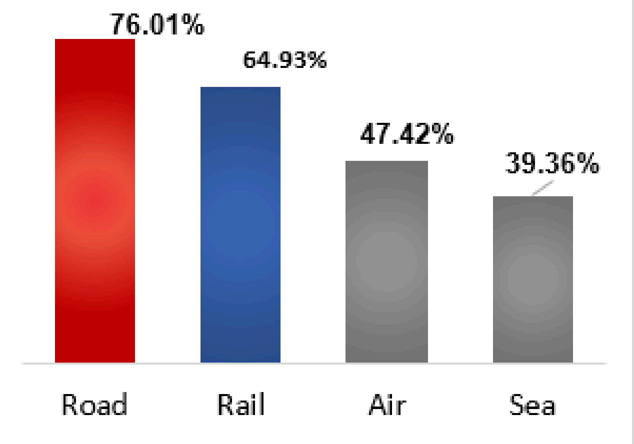
Profit Margin by Regional Hub



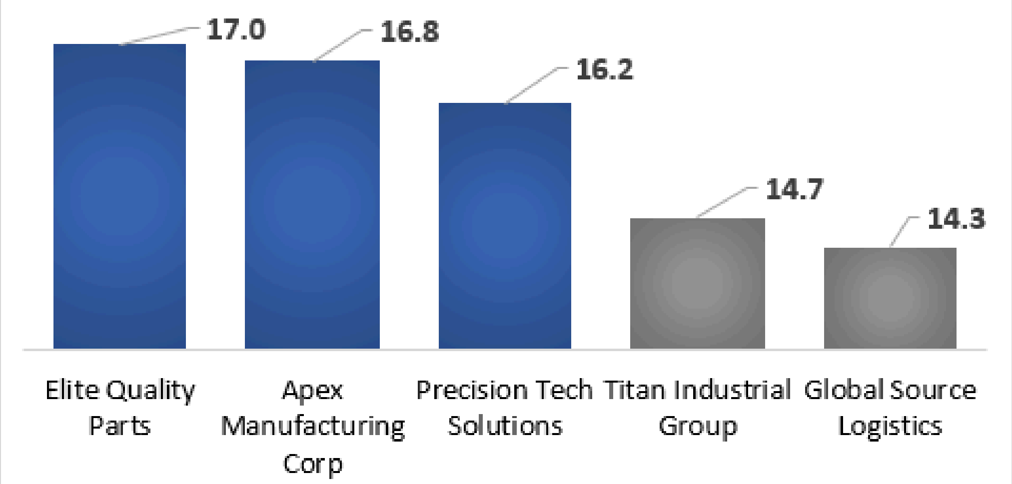
Shippping Cost Vs Speed



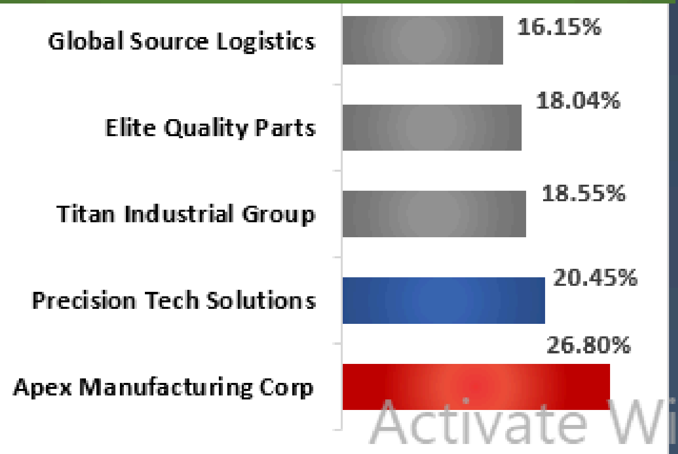
Defect Rate by Transport Mode



Avg. Lead Time by Supplier | in days



Defect Rate by Supplier



Strategic Summary

Top Margin

West Coast Hub leads at 57% Replicate their logistic model.

Risk Alert

Road transport causes 76% of defects. Shift high-value stock to Rail.

Lead Time

Audit Elite Quality Parts to reduc current 17day delays.

Inventory

Liquidate "Zombie" stock to boost cash flow.

A/B Test Intelligence | Ad Impact Analysis

Strategic Recommendations

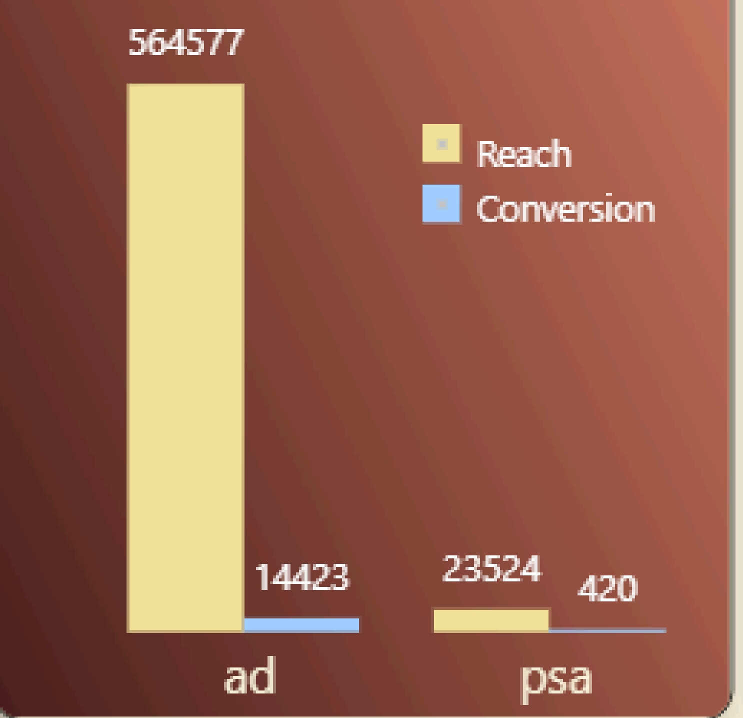
- **Optimize Spend** - Shift 20% of low-performing weekend budget to Monday Afternoon slots.
- **Capping Rules:** - Imple ment a 100-impression frequency cap to prevent Ad Fatigue and save budget.
- **Scale Winner** - Fully roll out the "Ad" creative as it proved a 43% lift over the baseline.
- **Content Pivot** - Redesign Saturday ad creatives to address the current weekly performance low.
- **Retargeting** - Focus follow-up budgets on users within the 20–80 impression sweet spot.

2.52%	2.55%	1.79%	43.09%
Overall Conversion	AD Success Rate	Baseline PSA	Total Campaign Lift

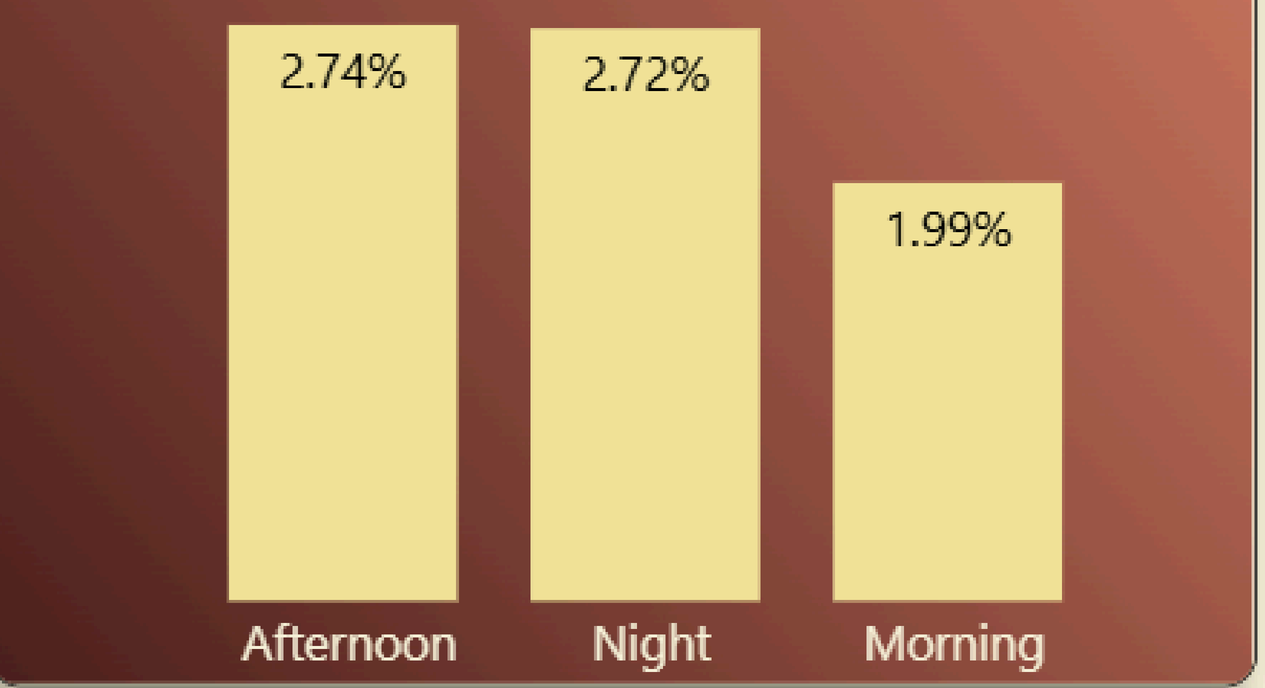
Test Group

ad | psa

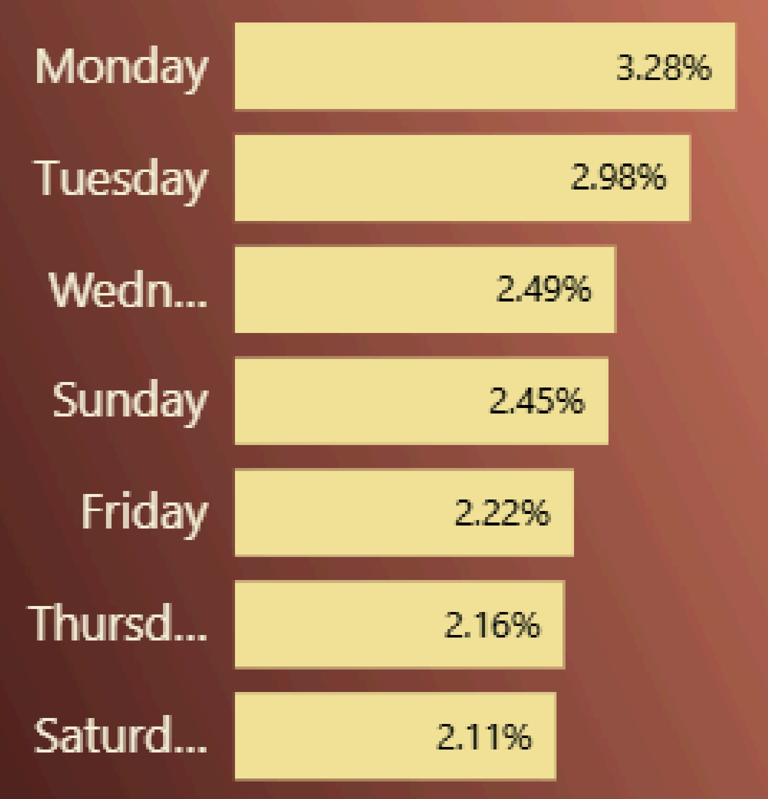
Reach Vs. Conversion



Time Block Performance



Daily Conversion Peak



Advertising Frequency Response

Bank Customer Churn Analysis Report

Maria Aslam

2026-01-26

Executive Summary

Strategic Overview

This report analyzes customer attrition across our European banking operations. While France and Spain maintain stable churn rates (~16%), the bank faces a critical emergency in the German market, where churn has spiked to 32.4%. This analysis shifts focus from *who is leaving* to *why our most valuable segments are exiting*.

Key Findings

- **TheTwo-Product Anchor:** We have identified a Survival Sweet Spot where customers with exactly 2 products have the highest loyalty (11–13% churn).
- **Product Over-Saturation:** A catastrophic failure occurs when German customers reach 3 or 4 products, resulting in a 90–100% exit rate. This suggests our complex bundles are driving customers away rather than locking them in.
- **Premium Segment Flight:** Churn is not limited to low-value accounts. High-balance customers and those in the “Very Good” credit tier (~33.5% churn) are leaving at nearly the same rate as high-risk segments.
- **Service Failure:** There is a near-total correlation between complaints and attrition. Currently, a complaint is a “guaranteed exit signal,” indicating our resolution process is failing to recover dissatisfied users.

Core Recommendations

- **Plus-One Retention:**Focusmarketing on moving 1-product holders to 2 products; avoid pushing to 3+ until the *Product Fatigue* issue is resolved.
- **German Priority Desk:** Launch a localized retention program for high-credit/high-balance German users to counter aggressive competitor poaching.
- **Active Recovery:** Implement a proactive “Re-engagement” protocol for inactive German users before they hit the 41.1% churn threshold

Data Cleaning & Methodology

To ensure the integrity of this analysis, the raw dataset under went a rigorous preprocessing phase within the R environment. This ensures that all statistical summaries and visualizations reflect accurate customer behavior.

The Cleaning Process

- **DimensionalityReduction:** Irrelevant features such as RowNumber, CustomerId, and Surname were removed. These variables contain unique identifiers that do not contribute to churn patterns and would introduce "noise" into the analysis.
- **Attribute Re-classification:** Several columns were imported as integers but represent categorical states. I converted Exited, HasCrCard, IsActiveMember, and Complain into Factors. This allows the analysis to treat them as distinct groups rather than continuous numerical values.
- **Data Consistency Check:** I verified the Geography column for missing values and spelling inconsistencies to ensure the reliability of the regional "Deep Dive" results.
- **Metric Calculation Logic:** During the summarization of churn rates, a nested as.numeric(as.character()) transformation was used. This ensures that binary factor levels are correctly interpreted for accurate percentage calculations.

Exploratory Data Analyis

Thedatasetcontains customer demographic, financial, and behavioral information used to analyze churn patterns. **Total customers** ≈ 10,000
Countries:

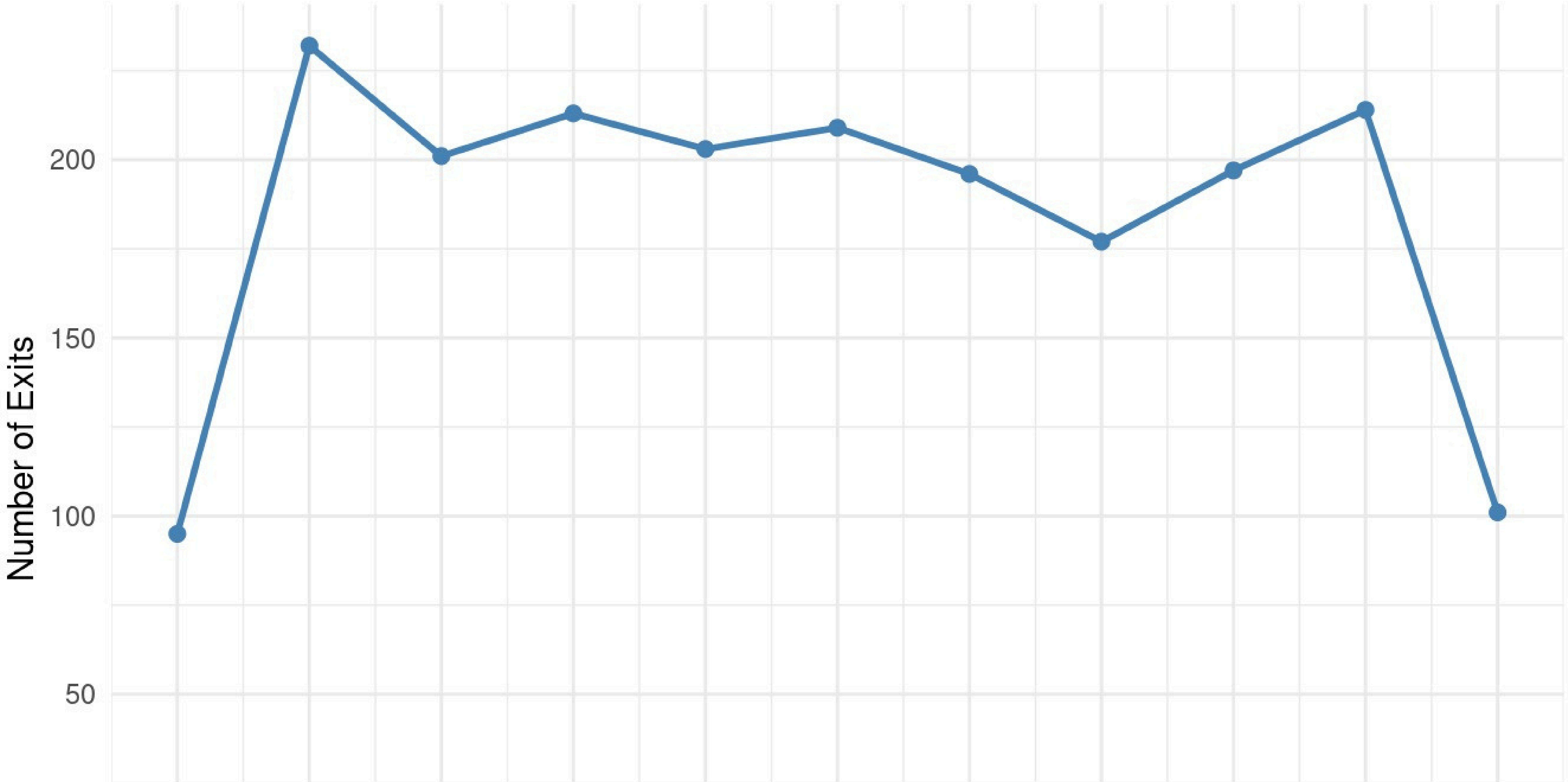
- France
- Germany
- Spain

Quick Summary

Total Customers	Total Exited	Active Members	Total Complaints	Churn Rate
10000	2038	5151	2044	20%

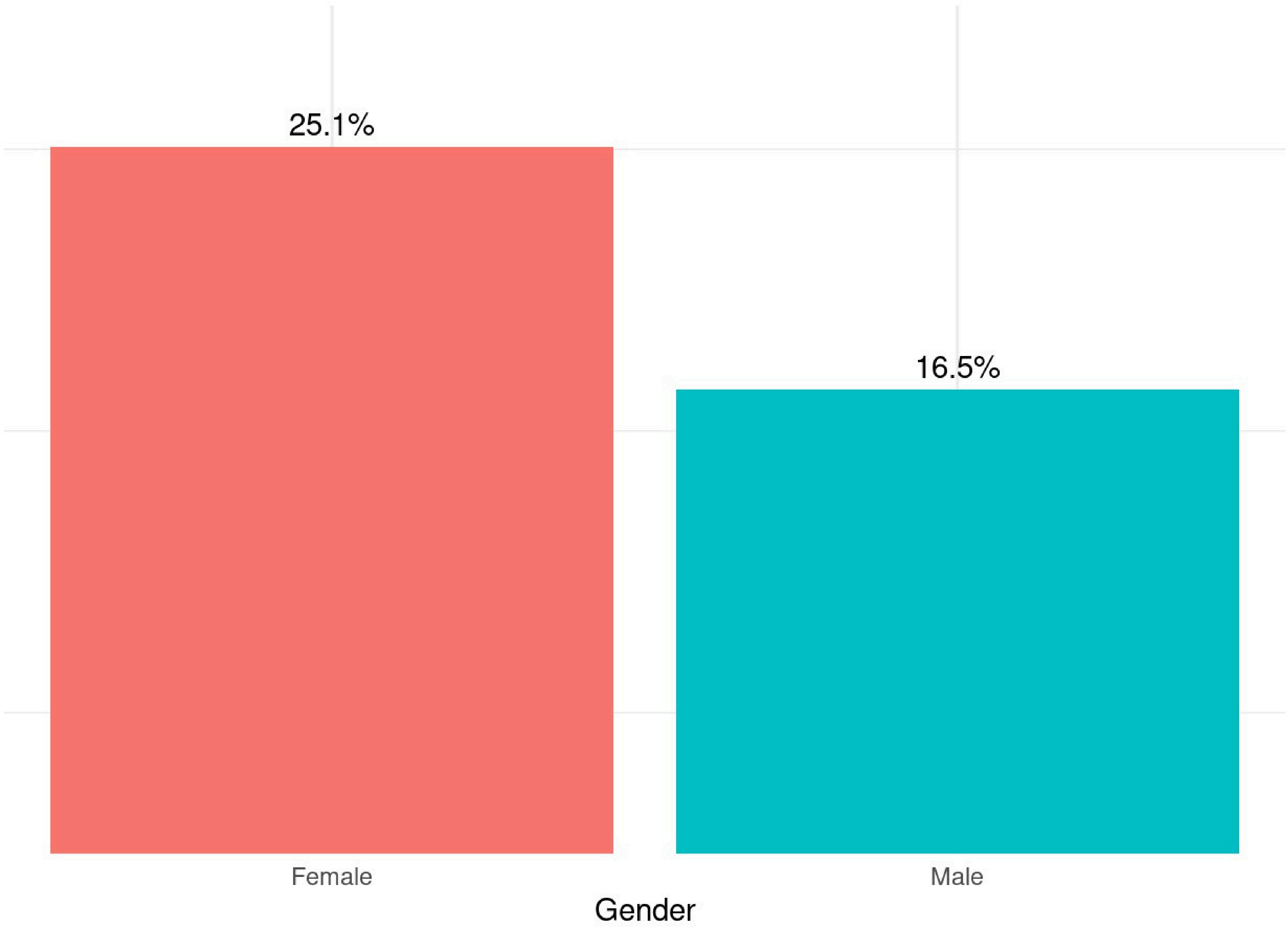
Approximately **20% of customers have churned** ,indicating moderate customer attrition risk.

Tenure Analysis



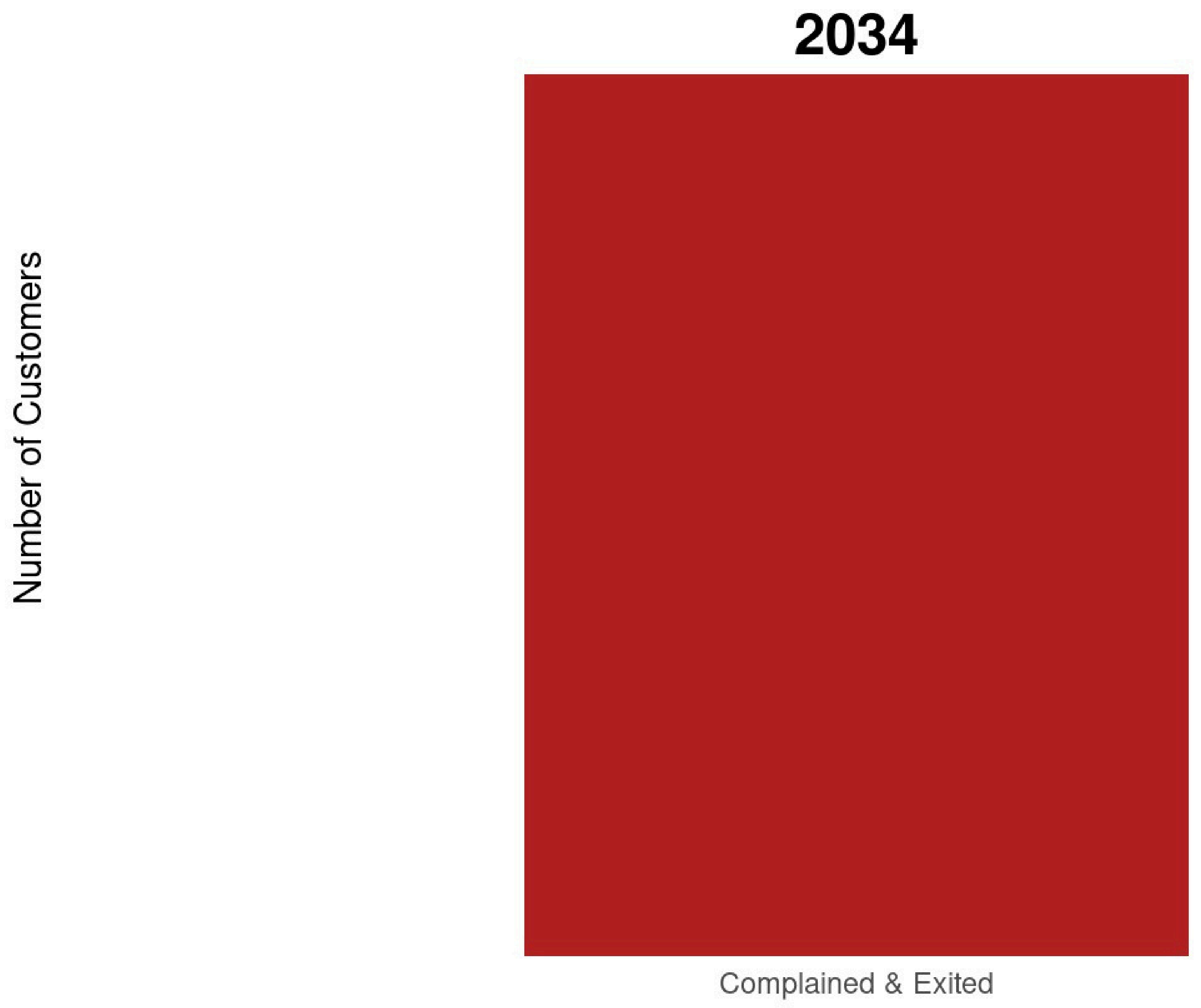
The data contradicts the standard **loyalty curve** where churn should decrease as tenure increases. To improve retention, the focus must shift from just winning back new joiners to addressing the **sustained dissatisfaction** that causes veteran customers of 5+ years to leave at the same rate as those of 0-2 years.

Churn by Gender

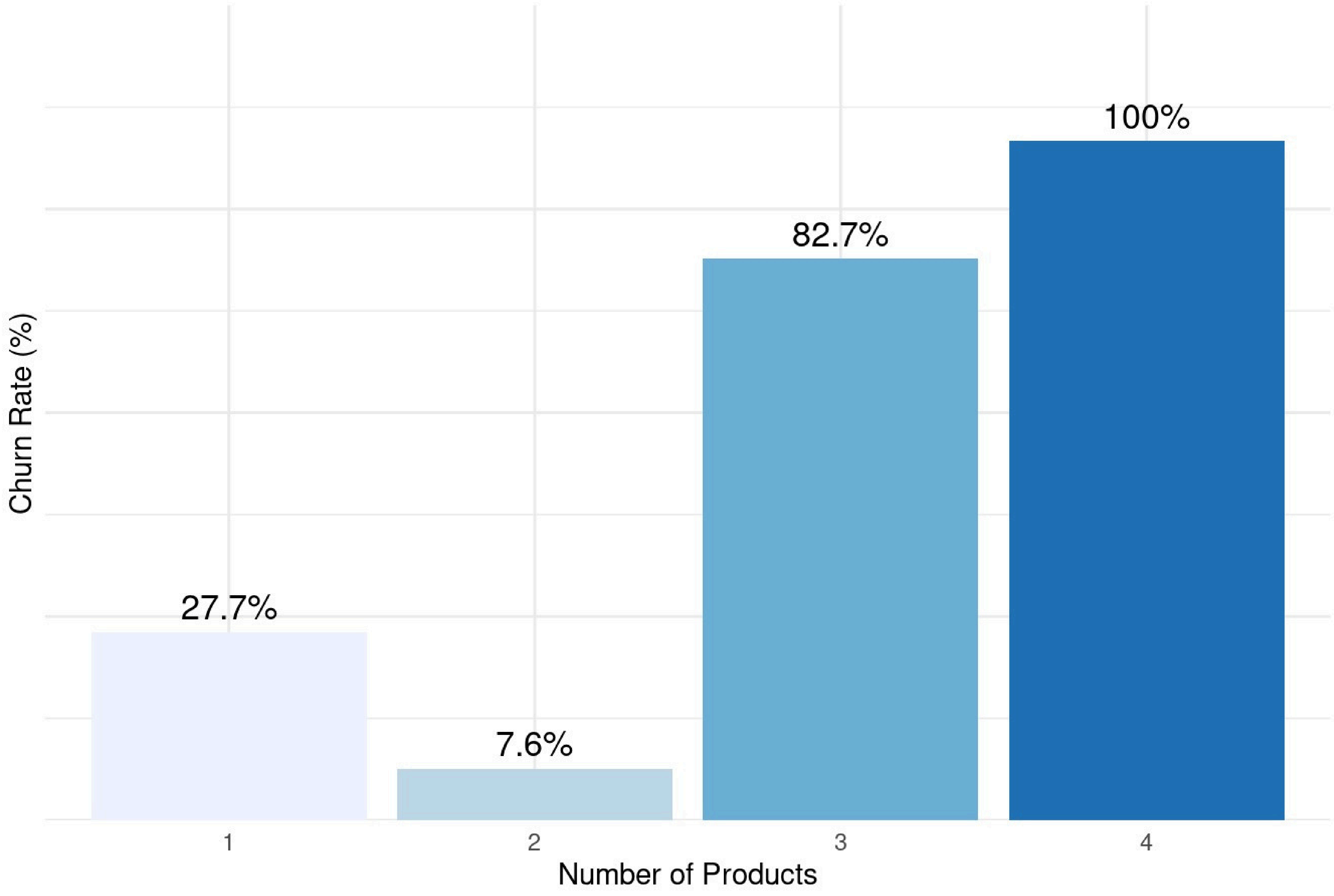


There is a stark contrast in retention across gender lines: Female customers exhibit a 25.1% churn rate, while Male customers churn at 16.5%. This represents an **8.6% absolute difference**, meaning women are roughly **1.5 times** more likely to leave the bank than men.

Complain Impact On Churn

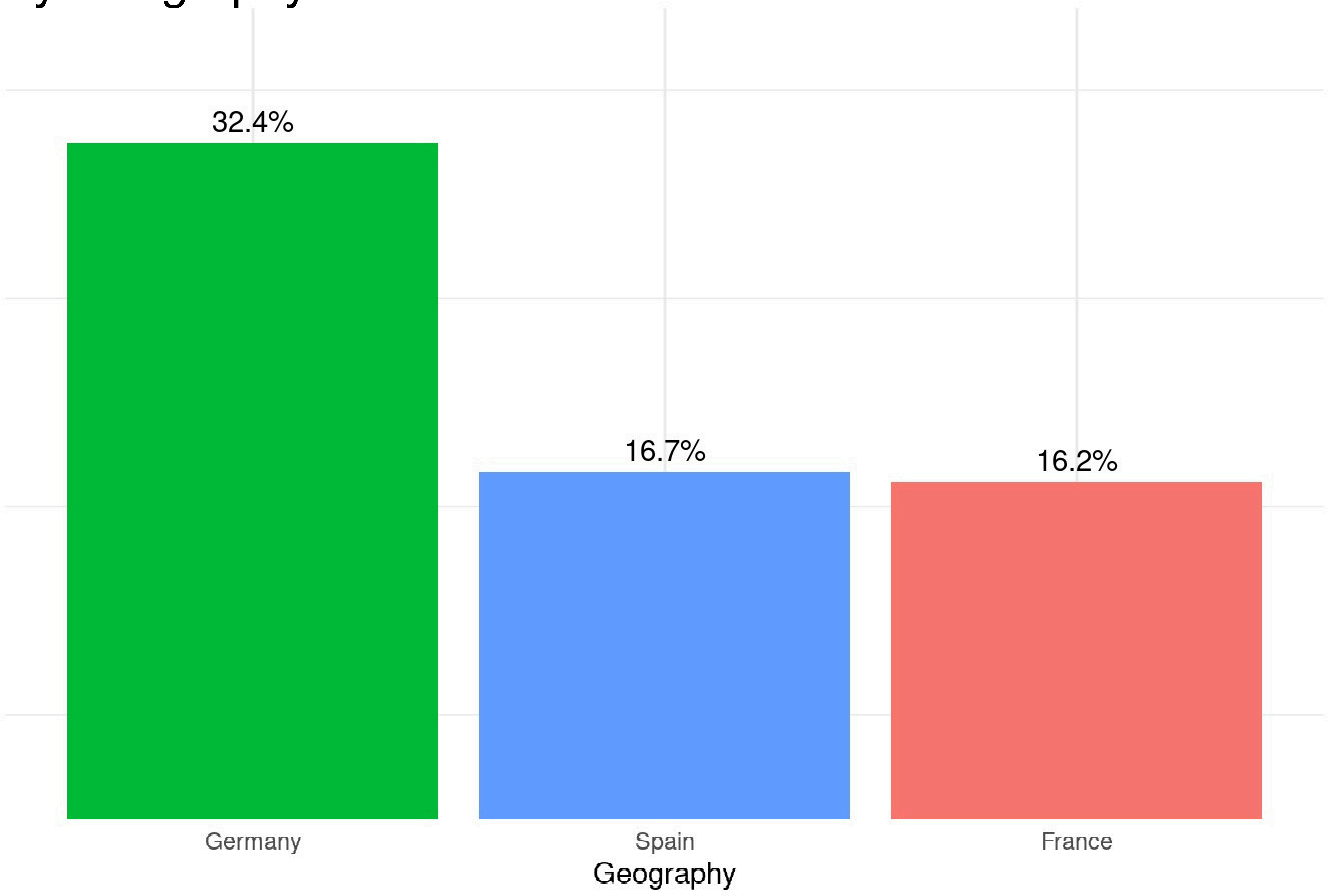


Product Vs Churn



The data reveals a *U-Shaped risk curve*. Customers with 2 products have the highest loyalty (lowest churn). However, churn increases significantly for Single-Product customers and **spikes dangerously** for those with 3 or 4 products (often exceeding (80-90%)

Churn by Geography



Germany requires an immediate **Deep Dive Audit**. We need to look at the German segment’s demographics, average account balances, and product holdings to see if we are losing our most profitable customers or if this is a general exodus

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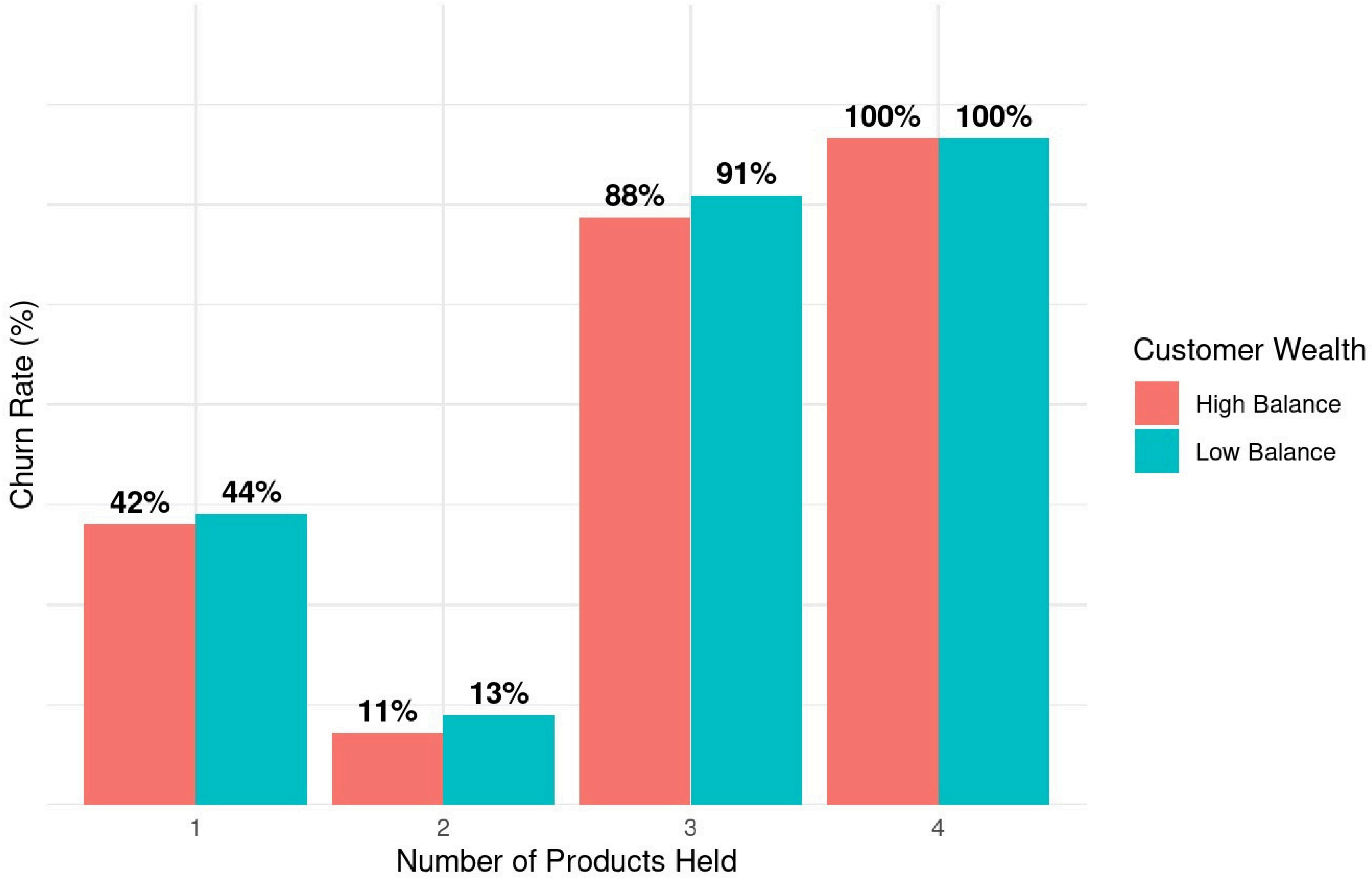
Deep Dive Analysis | Germany

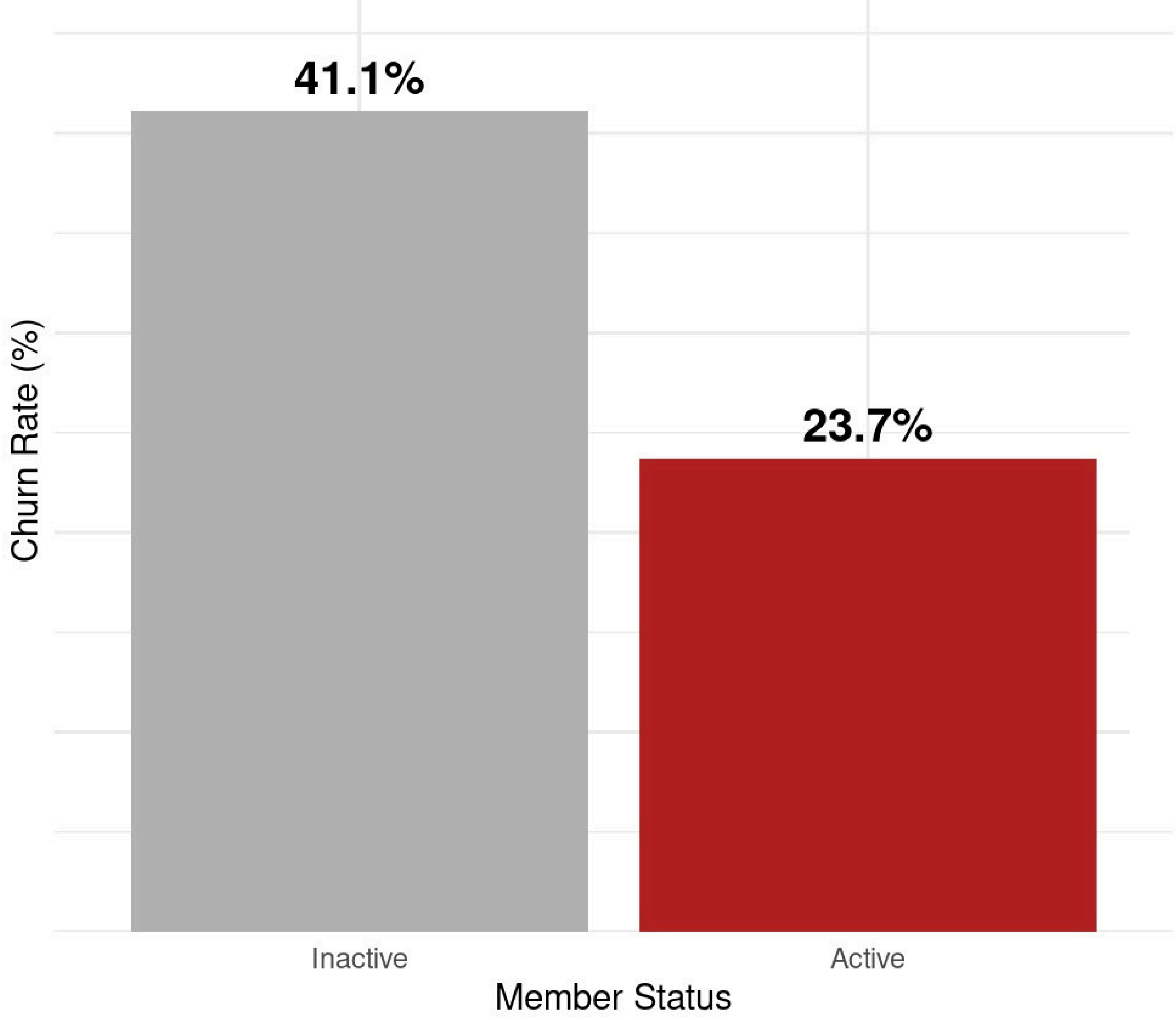
Analysis On Why is Germany’s Churn Rate Significantly Higher ?

Does Account Balance Influence Churn in Germany?

Notice that the High Balance and Low Balance are almost the same height in every category. This means German customers aren’t leaving because they are rich or poor; they are leaving based on how many products they hold

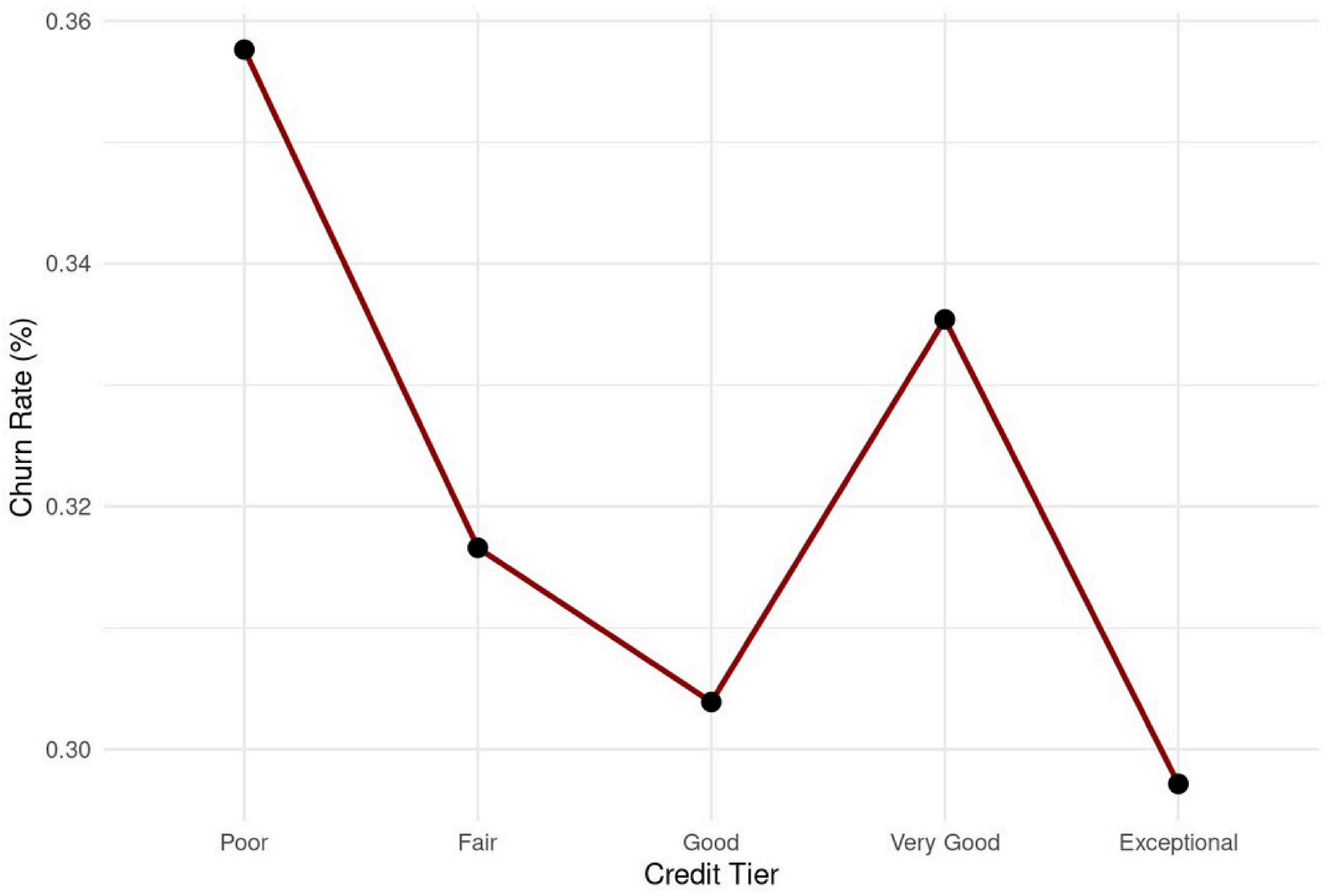


Can Customer Engagement Help Reduce Churn in Germany?



Activity is a major factor in retention, but it is **not a cure**. Inactive members in Germany are in a state of collapse with a 41.1% churn rate, while Active members still churn at 23.7%

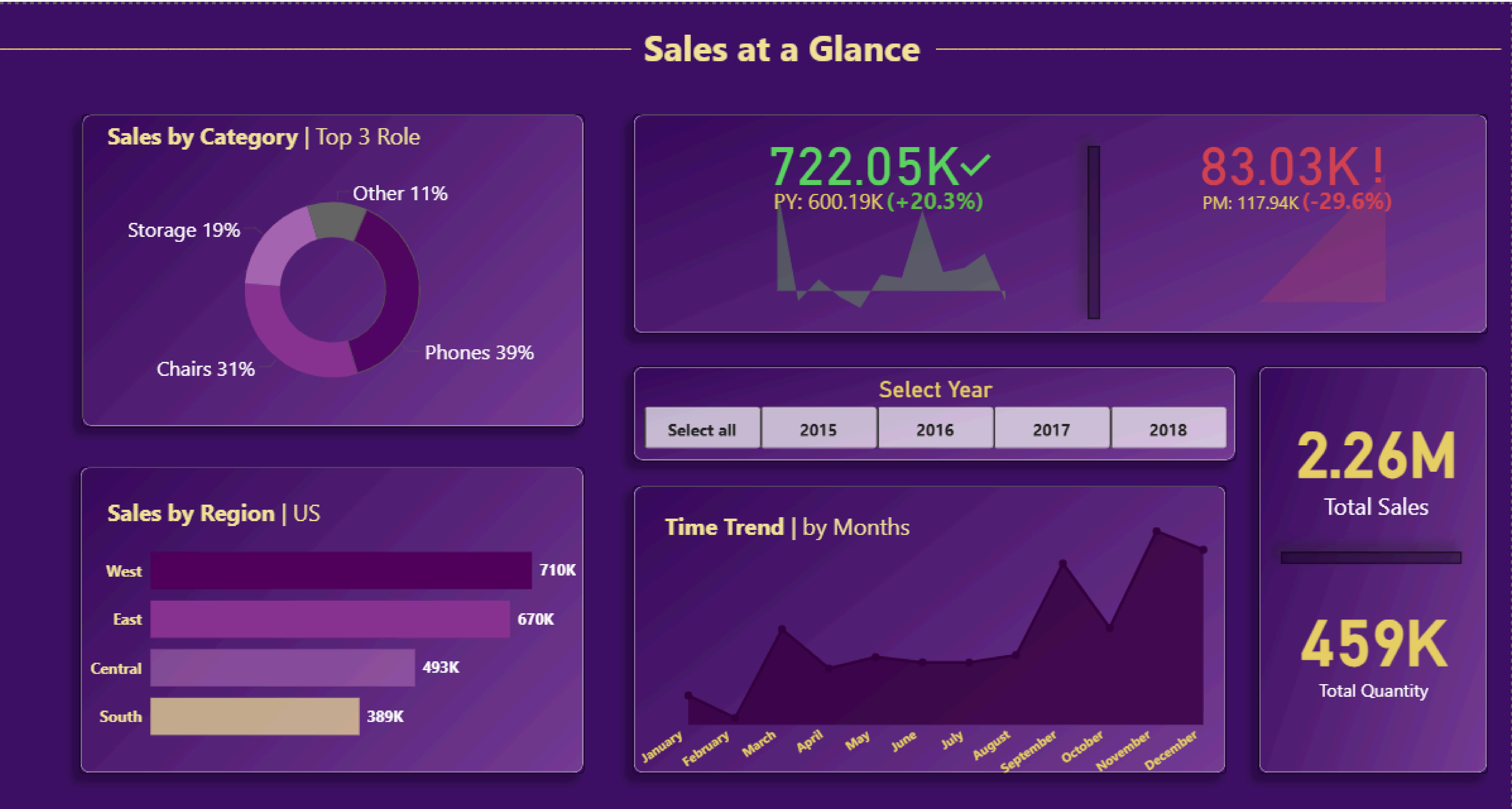
How Credit Quality Affects Customer Churn in Germany?



Highest churn is among *Poor* credit holders 36%,there is a significant and concerning spike in the ***Very Good tier*** **(33.5%)**

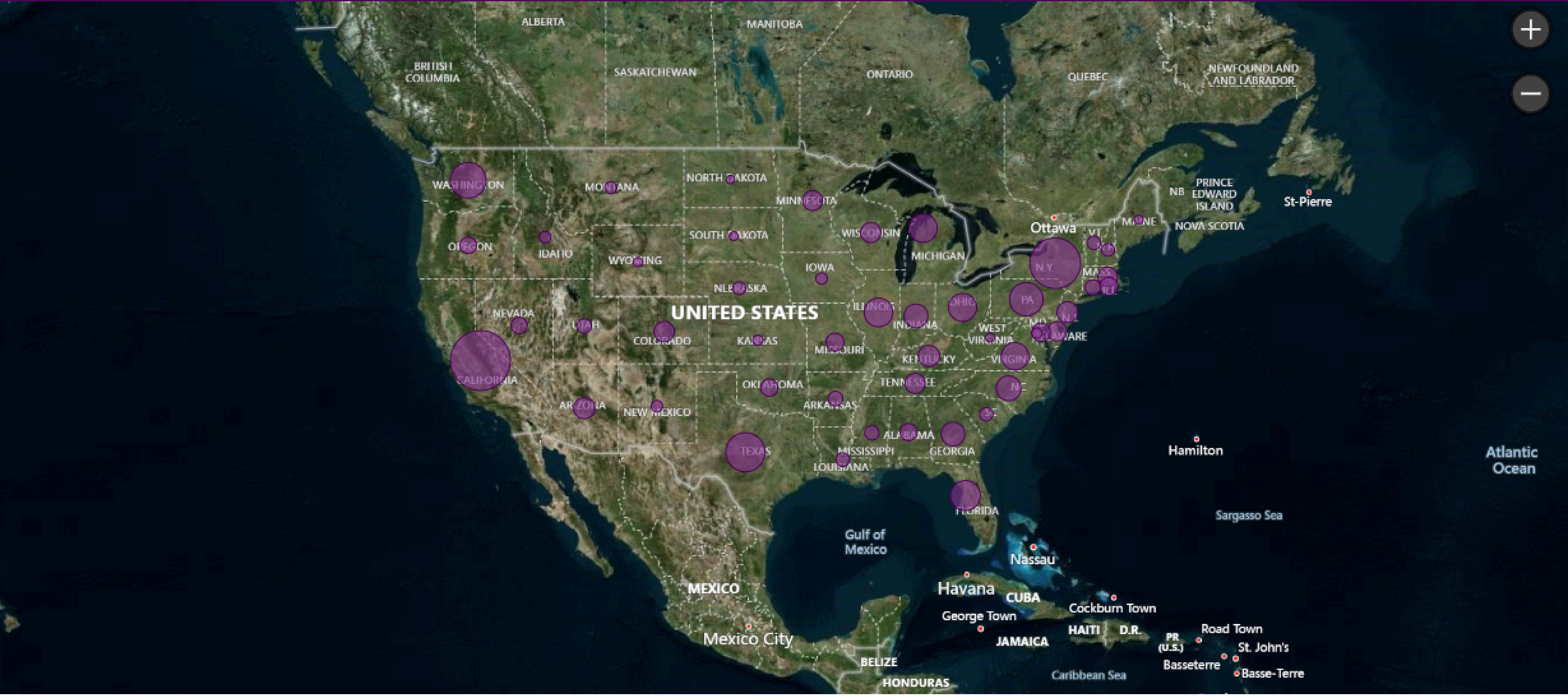
Strategic Conclusion

The data confirms that the bank's issue in Germany is not a lack of customer engagement, but a **gap in premium relationship management**. We are successfully attracting high-value, active, and credit-worthy customers, but our current ecosystem—specifically our complaint resolution and multi-product bundling—is creating *friction* that drives them to competitors. By stabilizing the German market through a *Plus-One* product strategy and a dedicated complaint-recovery desk, we can protect the bank's most profitable assets and reduce regional churn back to sustainable levels.


Sales at a Glance
Sales by Category | Top 3 Role
Other 11%
Storage 19%
Chairs 31%
Phones 39%
722.05K
PY: 600.19K (+20.3%)
83.03K !
PM: 117.94K (-29.6%)
Select Year
Select all
2015
2016
2017
2018
2.26M
Total Sales
459K
Total Quantity
Sales by Region | US
West
710K
East
670K
Central
493K
South
389K
Time Trend | by Months
January
February
March
April
May
June
July
August
September
October
November
December

Select Region

Select all | Central | East | South | West



Marketing Campaign Performance & ROI Audit

Maria Aslam
Janurary 2026

Executive Summary

Strategic Overview

This report evaluates the performance of **$≈ 62.37 Million** in total marketing expenditure over a rolling 12-month period. While the portfolio maintains a stable **6.01% Weighted ROI**, the analysis reveals a Performance Ceiling caused by top-of-funnel engagement gaps. This summary shifts focus from general channel maintenance to creative optimization as the primary driver for future growth.

Key Findings

- **The Stability Paradox:** ROI remains virtually identical across all channels and customer segments. This indicates a mature, low-risk portfolio but suggests that budget reallocation alone will not generate a performance breakthrough.
- **Top-of-Funnel Bottleneck:** We are achieving massive scale in impressions, but facing a significant drop-off to Clicks (8.5% CTR). The current strategy is successful at reaching people but failing to "hook" them at the first touchpoint.
- **Ad Fatigue Resilience:** Analysis of campaign duration (5–30 days) shows zero ROI decay. Our creative content is "evergreen," meaning we can run longer campaign flights without losing profitability.
- **Cost-Driven Volatility:** The ROI dip observed in April was not a result of falling consumer demand, but rather minor fluctuations in acquisition costs. Our margins are more sensitive to internal spend efficiency than external market shifts.

Core Recommendations

- **Creative Pivot (CTR Focus):** Shift 100% of A/B testing efforts toward *Thumb-Stopping* creative content to improve the **8.5% CTR.**
- **Evergreen Deployment:** Transition to longer campaign durations (21+ days) to reduce operational overhead, as performance does not drop over time.
- **Efficiency Buffer:** Implement a Cost-Cap protocol during the Q2 period to prevent the minor spend spikes that previously eroded our ROI margins.

Data Cleaning & Methodology

Toensurethe integrityofthis analysis, therawmarketing datasetunderwent a rigorous preprocessing phase. This ensures that all statistical summaries and visualizations reflect accurate campaign performance and customer engagement metrics.

The Cleaning Process

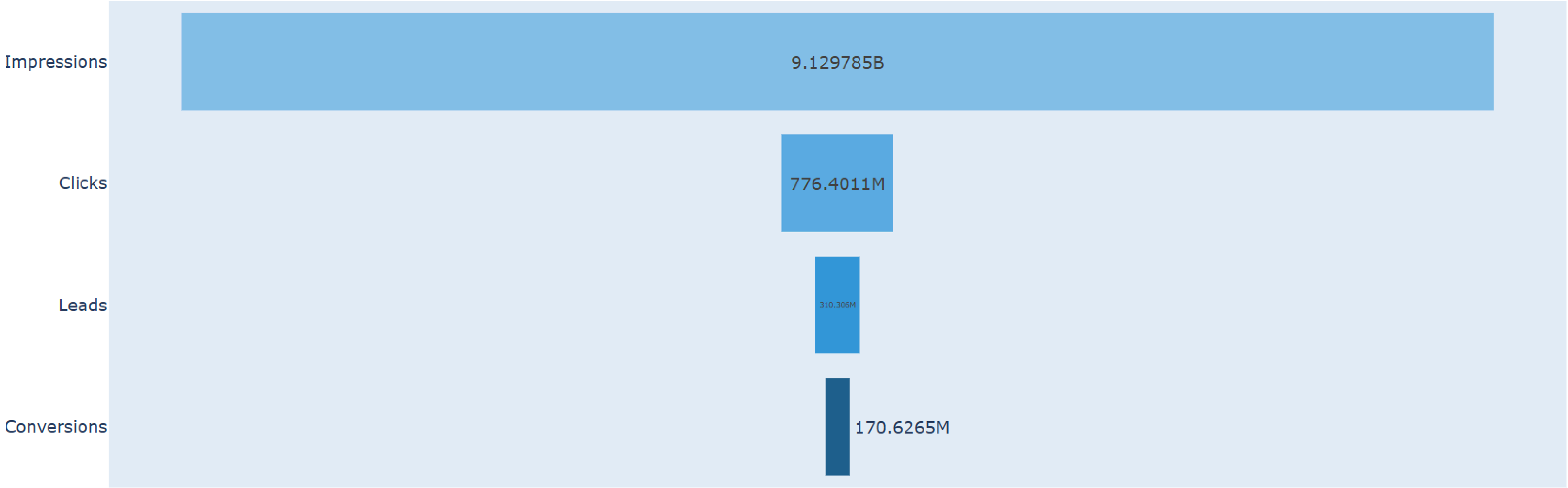
- **Duplicate & Integrity Check:** I performed a comprehensive scan for duplicate entries and verified that each campaign record was unique. This prevents the artificial inflation of revenue and cost metrics.
- **Missing Value Strategy:** I audited the dataset for null values across critical columns like Revenue and Acquisition_Cost. Any inconsistencies were resolved to ensure a 100% complete data profile for the $62.3M spend analysis.
- **Type Casting & Formatting:** Converted the Date column into a datetime64 object to enable precise chronological sorting and seasonal trend analysis.
- **Dimensionality Optimization:** Removed non-essential system identifiers and ensured that categorical features (such as Campaign_Type and Target_Audience) were properly encoded for grouping operations.
- **Statistical Validation:** Every calculated metric (ROI, Conversion Rate, CTR) was cross-referenced against raw sums to ensure the "Weighted" averages accurately represent the true financial impact.

Exploratory Data Analysis - EDA

Thisphase involvesamulti-dimensional examination of **$62.3M** inmarketing spend, cross-referencing campaign performance against localized audience behaviors and channel-specific trends. By segmenting the data through a 12-month lens, we identify the specific drivers of conversion and the mechanical bottlenecks within our sales funnel.

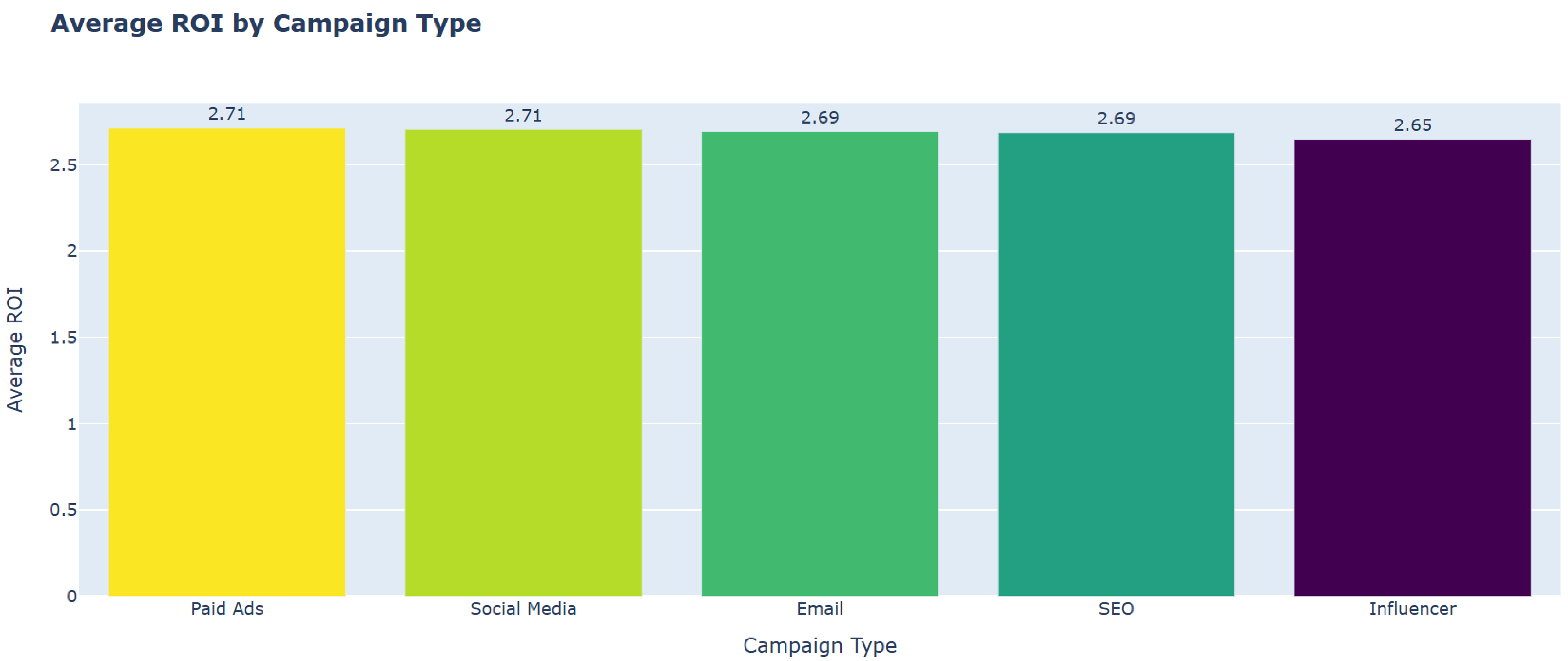
	Impressions	Clicks	Leads	Conversions	Revenue	Conv. Rate	Weighted ROI	Total Spend
Totals	9129.8M	776.4M	310.3M	170.6M	85207.5M	21.94%	6.01%	$62,365,216.56

Marketing Funnel Performance



> **Key Takeaway:** The funnel shows a significant drop-off from **Impressions 9.1B** to Clicks (776M), representing an **8.5% Click-Through Rate**. However, the conversion efficiency from Leads to Conversions remains strong, suggesting that while top-of-funnel reach is vast, audience targeting could be further refined to improve initial engagement where optimization strategies could improve customer conversion performance.

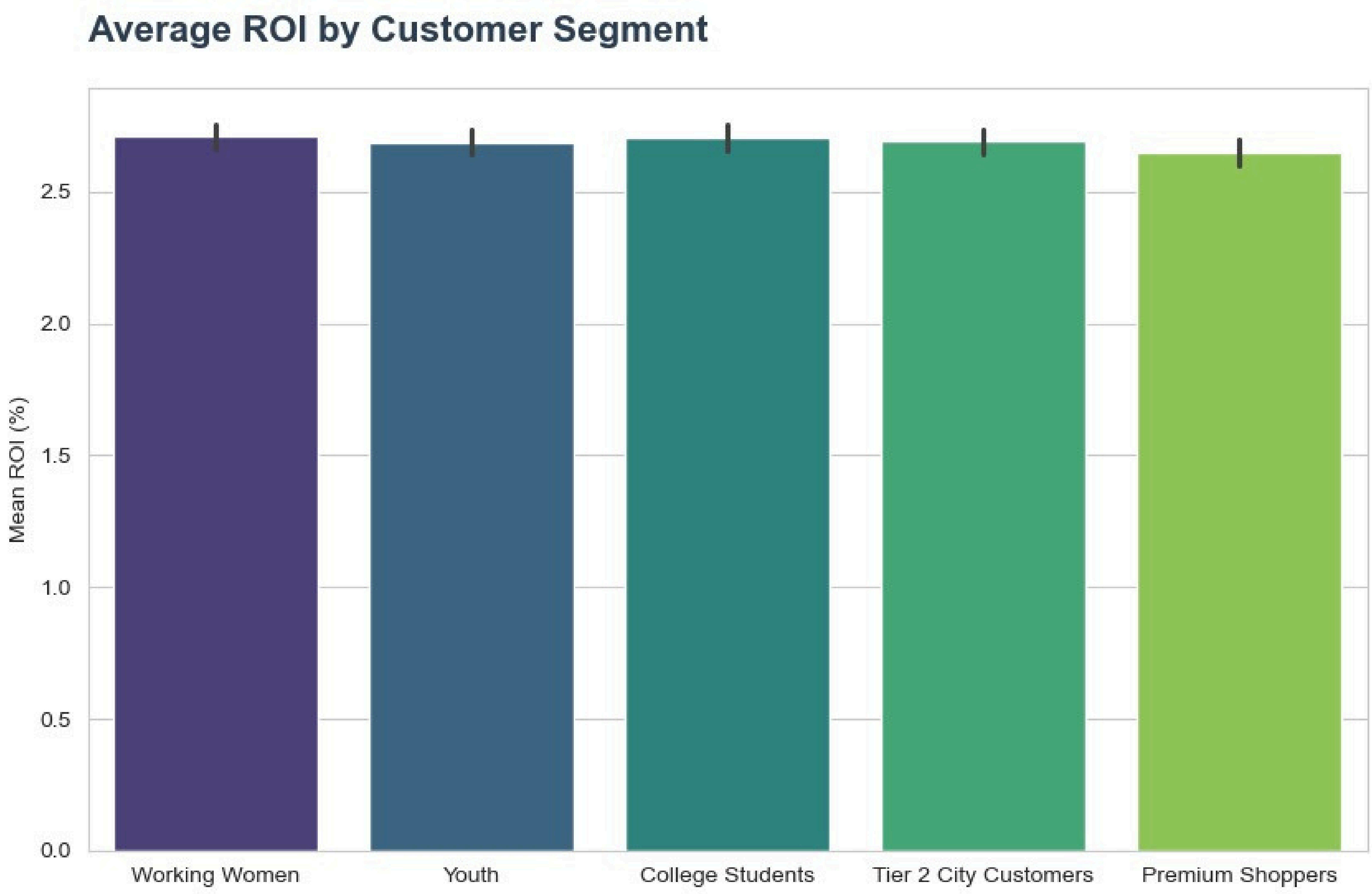
ROI Efficiency Analysis



> **Key Takeaway:** While ROI is consistent across all channels, this suggests that **efficiency is no longer the deciding factor** for budget allocation. Management should instead focus on **Volume and Scalability**, shifting funds toward channels with the highest total Revenue while maintaining this stable 6% ROI floor.

Analysis of Customer Segments

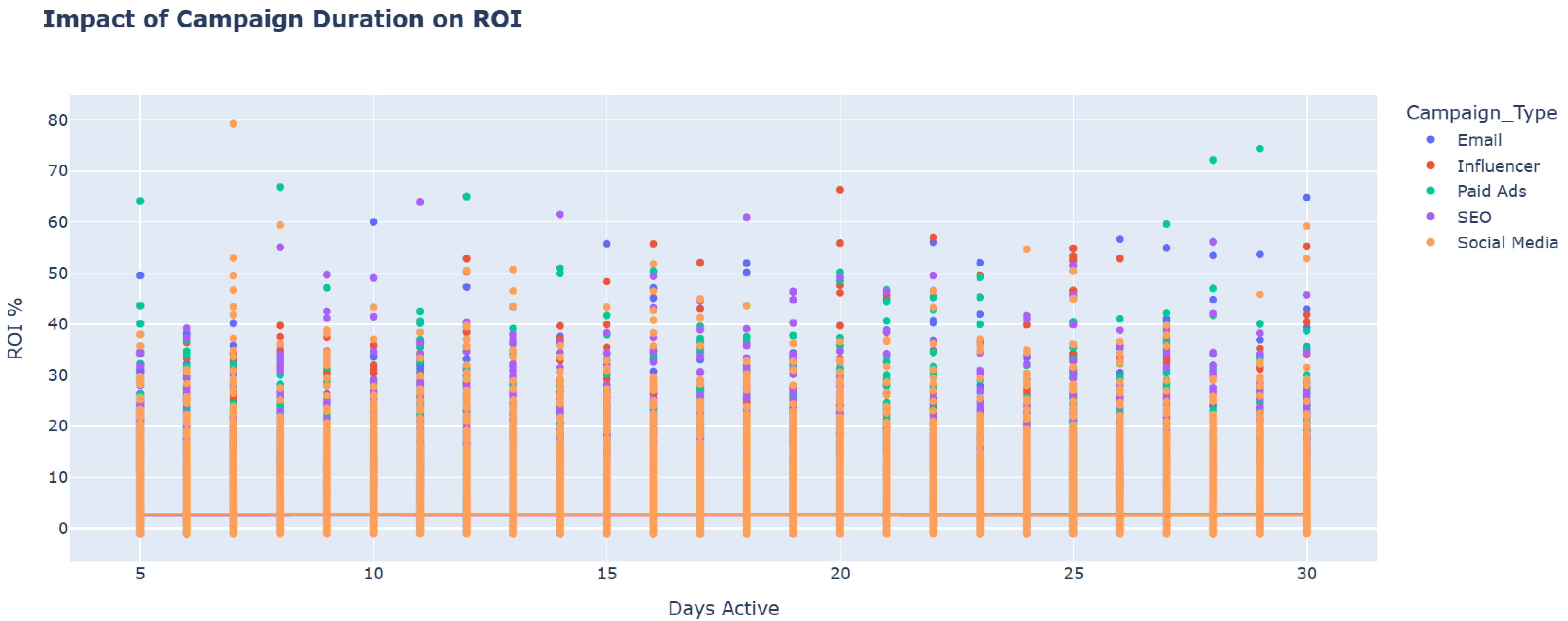
Average ROI across different customer segments to identify if specific groups respond better to our campaigns.



> **Analysis Insight:** The ROI remains consistent across all customer segments, further confirming that our marketing message resonates equally across the entire database. This indicates a **highly stable and predictable revenue model**, where no single demographic is being underserved or over-exploited.

Duration Impact Analysis

We analyzed the relationship between campaign length and profitability to identify "ad fatigue."



Analysis Insight: The flat trendline in the Duration vs. ROI analysis indicates zero diminishing returns. Campaigns maintain their efficiency regardless of length (up to 30 days). This allows for longer-term campaign planning without the risk of performance decay.

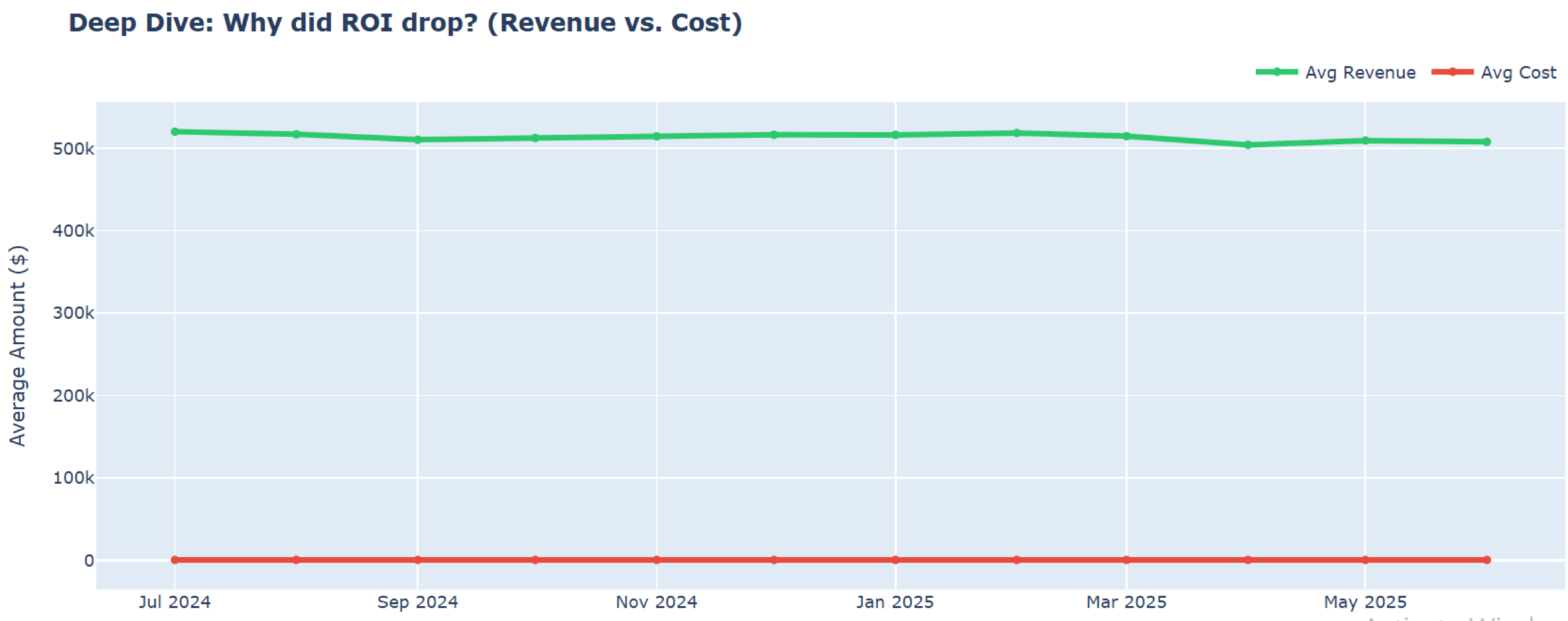
ROI Trend

By looking at the full timeline from 2024 through 2025, we can see how the marketing strategy has evolved over time rather than just looking at a single calendar year.

> **Key Takeaway:** The view reveals that ROI peaked in **February 2024** and has remained relatively stable since. The lack of significant "year-over-year" growth suggests the campaign has reached a steady state of performance, providing a predictable 2.7% return on investment.

Deep Dive: Root Cause Analysis of the Q2 ROI Dip

To move beyond surface-level trends, we analyzed the direct relationship between Revenue and Acquisition Cost. This identifies whether our April performance drop was caused by rising ad costs or a decrease in consumer purchase intent.



> **Root Cause Analysis:** The Deep Dive chart shows that **Revenue remained stable** even during the ROI dip. This suggests that the April decline was not due to a loss of customers, but rather a slight **reduction in operational efficiency**. Since costs and revenue are so far apart, even a minor $100 increase in cost can cause the ROI percentage to fluctuate.

Conclusion & Strategic Recommendations

Basedonthemulti-dimensionalanalysisofthemarketingdatafrom2024-2025, we have identified three core strategic pillars for the upcoming quarter.

Area of Focus	Key Finding	Recommended Action Prioritize budget for
Channel Strategy	Uniform ROI across all Campaign Types.	high-revenue channels like Social Media. Maintain current
Efficiency	Overall Weighted ROI stabilized at 6.01%.	spend levels; focus on cost reduction. A/B test ad creatives to
Growth Opportunity	Significant drop-off at the top of the funnel (8.5% CTR).	improve Click-Through Rate.

> The data confirms that the current marketing challenge is not a lack of scale or channel selection, but a bottleneck in engagement quality. We are successfully achieving massive reach and maintaining a high lead-to-sale conversion rate (21.94%), but the current ecosystem is **losing the vast majority of potential revenue at the Click-Through (CTR) stage.** Because our ROI remains stable over long durations without decay, we do not need more frequent campaign refreshes; rather, we need a higher creative hook to capture the billions of impressions currently being wasted. By focusing on Top-of-Funnel optimization and maintaining strict cost-caps during the Q2 period, we can maximize the efficiency of our $62.3M budget and push the portfolio beyond its current 6.01% ROI ceiling.

Final Statement

Themarketingportfolio iscurrently in a **Steady State** of performance. While risk is low due to consistent returns across all segments, the primary path to increased profitability is not through channel shifting, but through **optimizing the engagement funnel** to convert a higher percentage of our 9.1B impressions.

Coca.Cola (KO) Stock Market Performance | Oct 2019 - Oct 2022

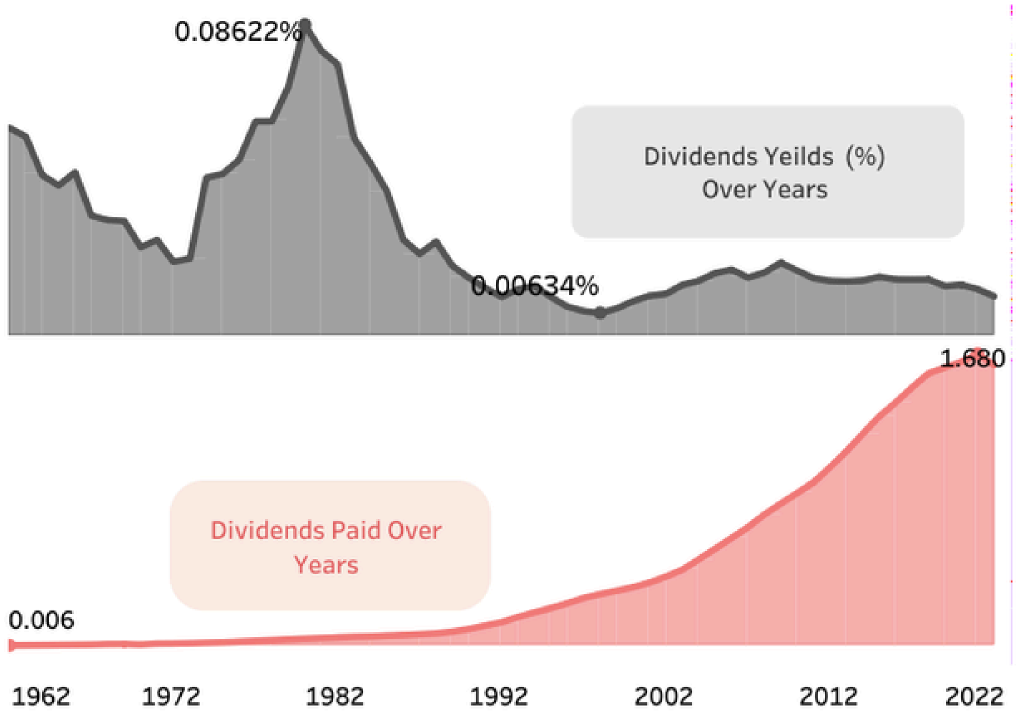
Total Volume
4,253M
▲ 8.08% PY (3,935M)

Max. Closing Price
$65.26
▲ 17.85% PY ($55.37)

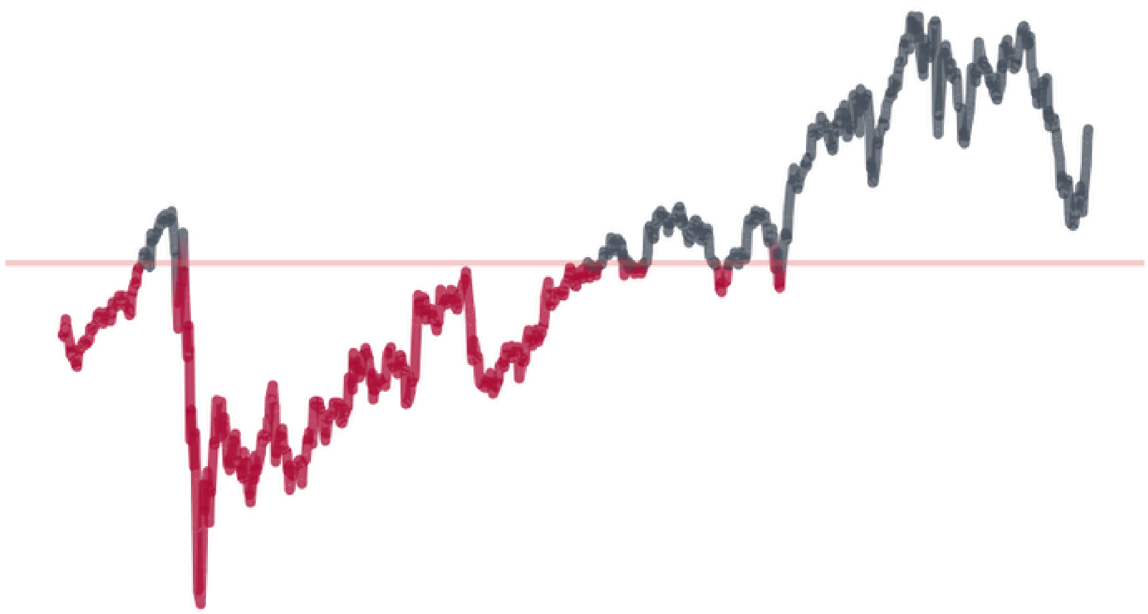
$1.02
▲ 51.22% PY ($0.68)

Export As

Follow



Avg. Closing Price | Oct 2019 - Oct 2022

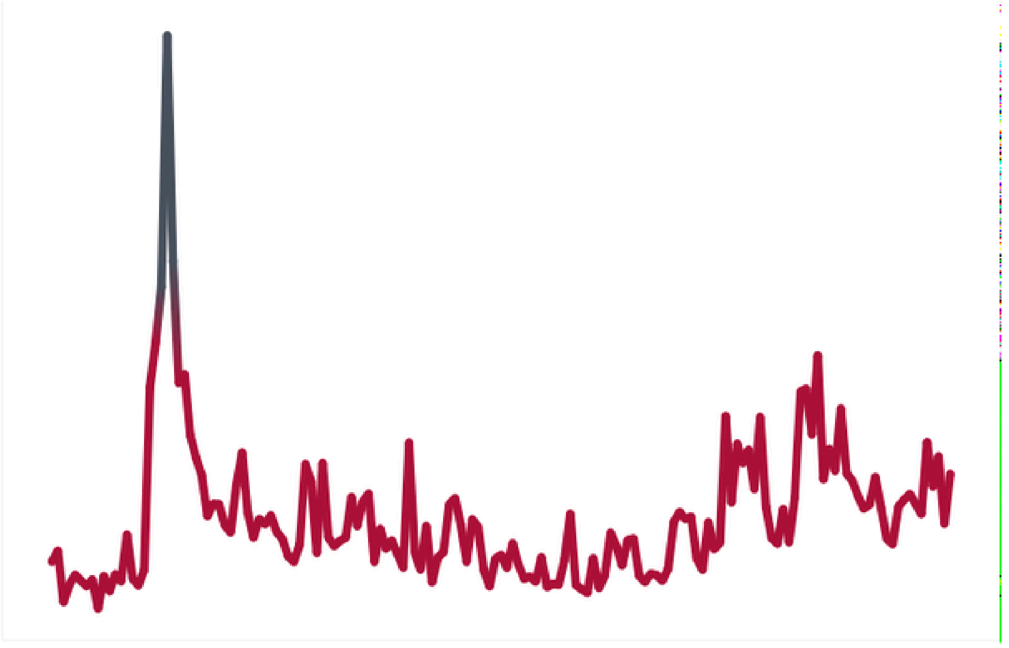


Highest Price
$66.24

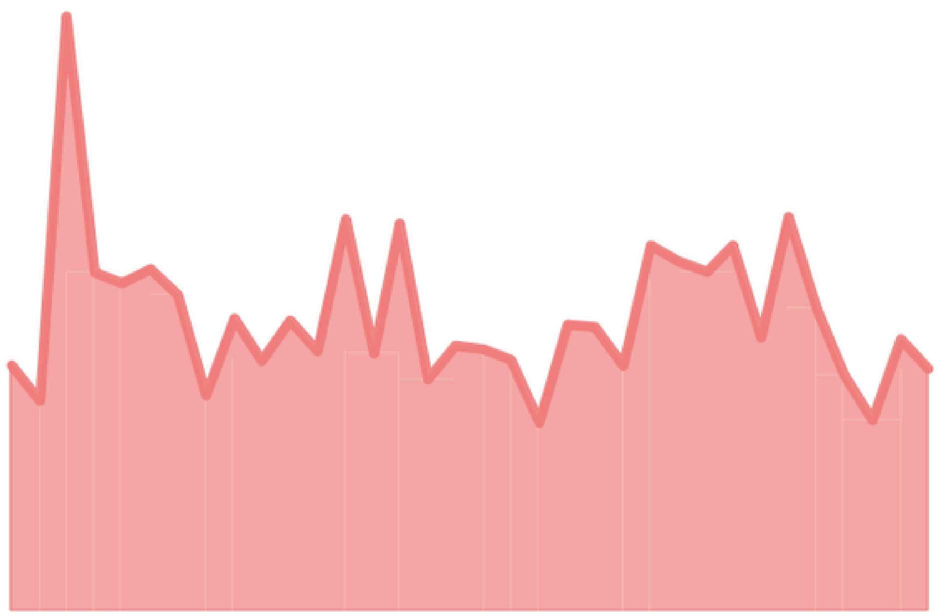
Closing Price
$52.40

Lowest Price
$33.55

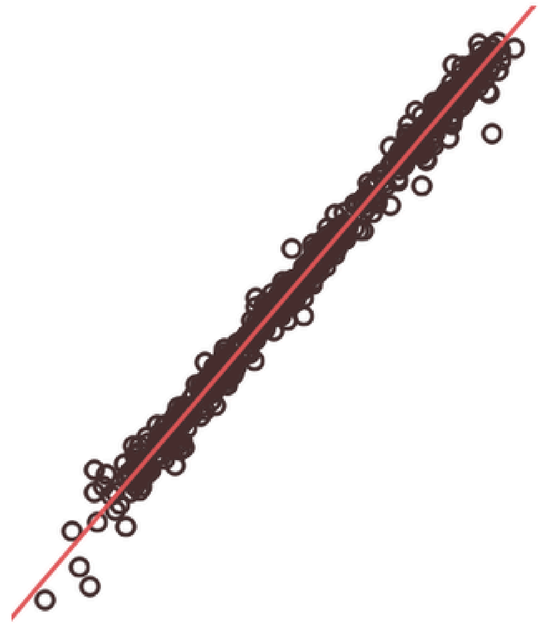
Stock Votality | Oct 2019 -Oct 2022



Sum of Volume | By Months



Open Vs Close | Daily Basis



Select Year

All

Sales Trends & Revenue Drivers

Exploring category-wise performance to highlight the backbone of total sales

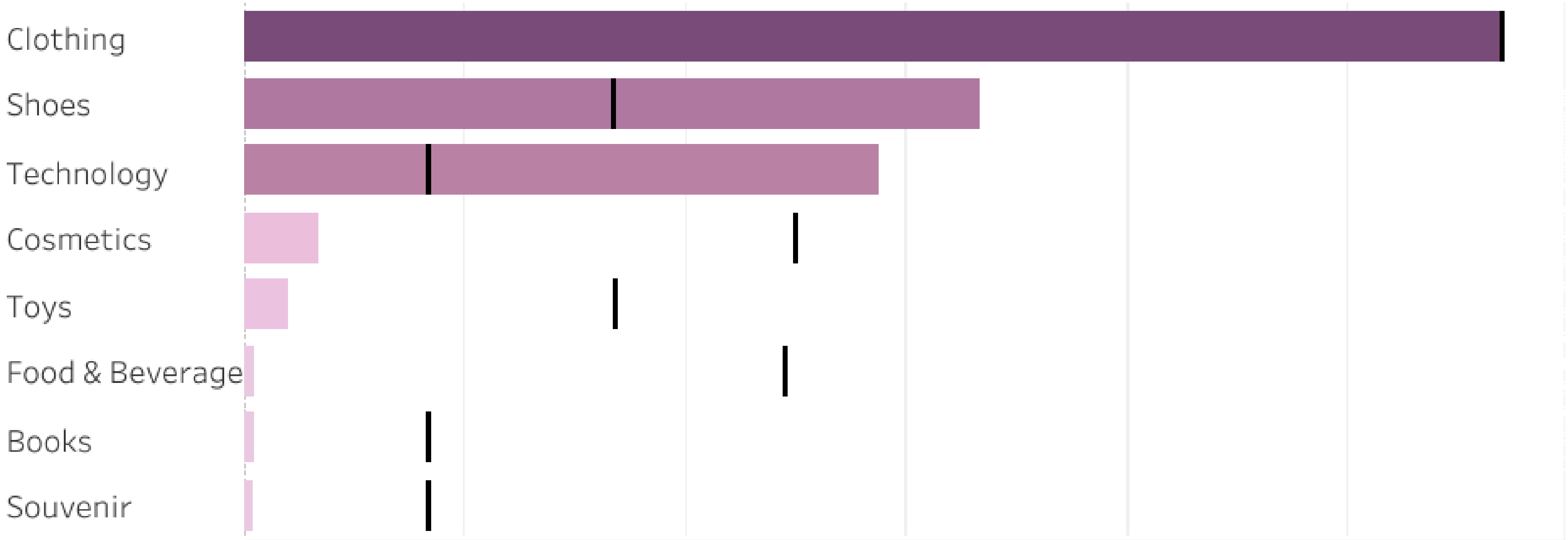
Navigate

Total Customers
90,933

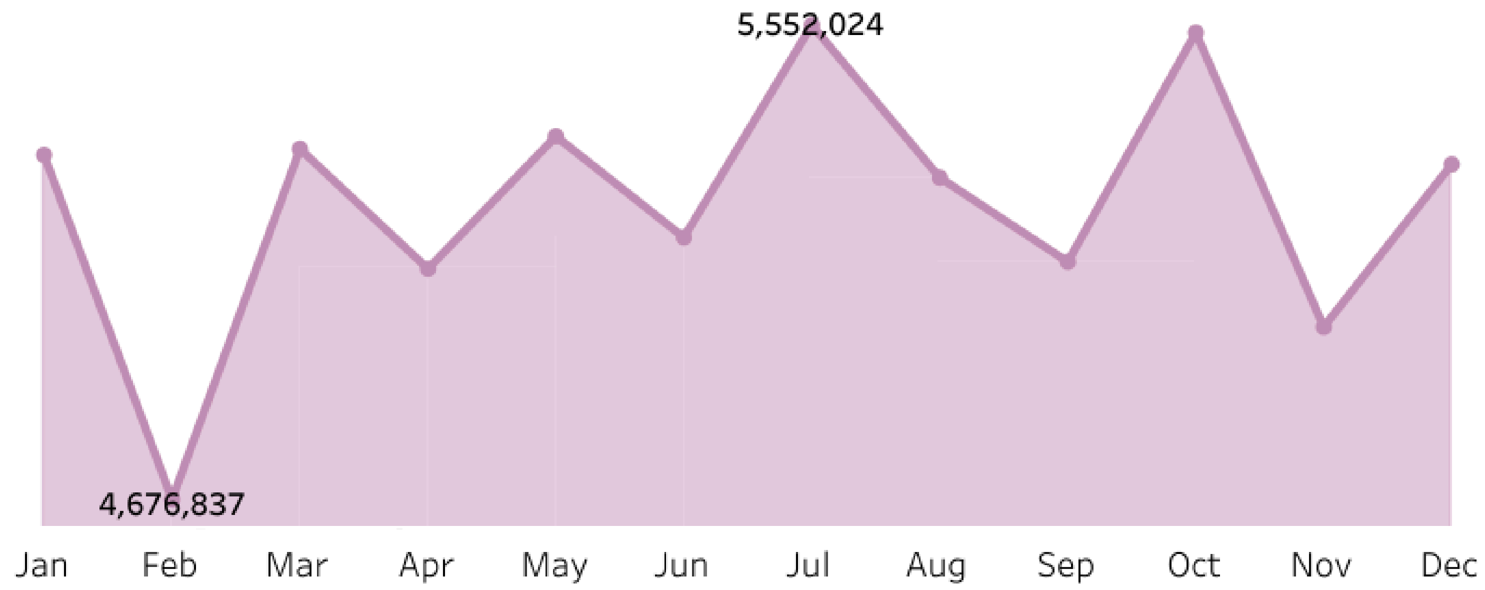
Total Sale Price
$63M

Total Quantity Sold
273K

Sales by Category | Vs Quantity Sold

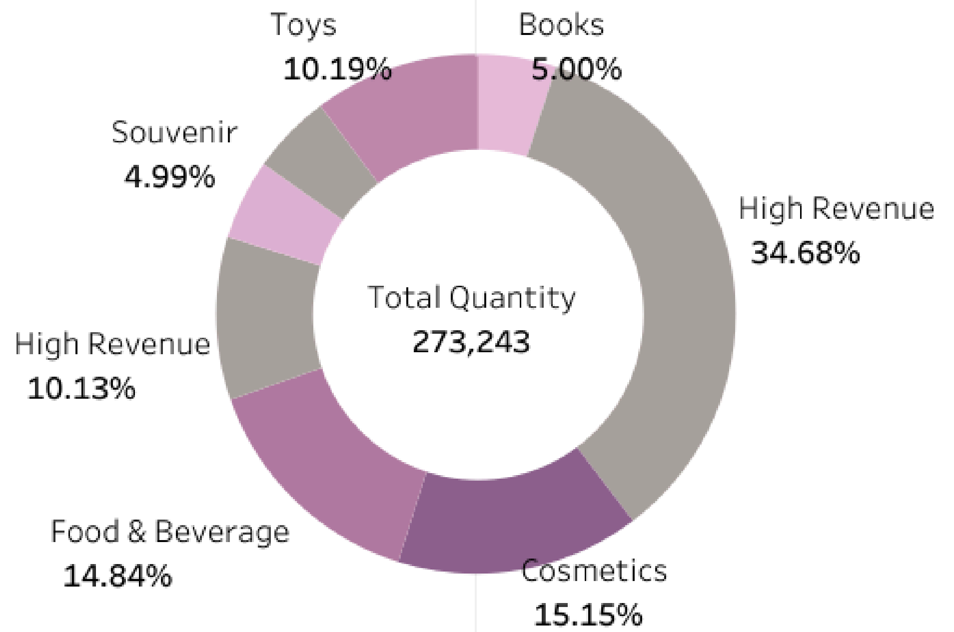


Time Trend | Monthly Trend

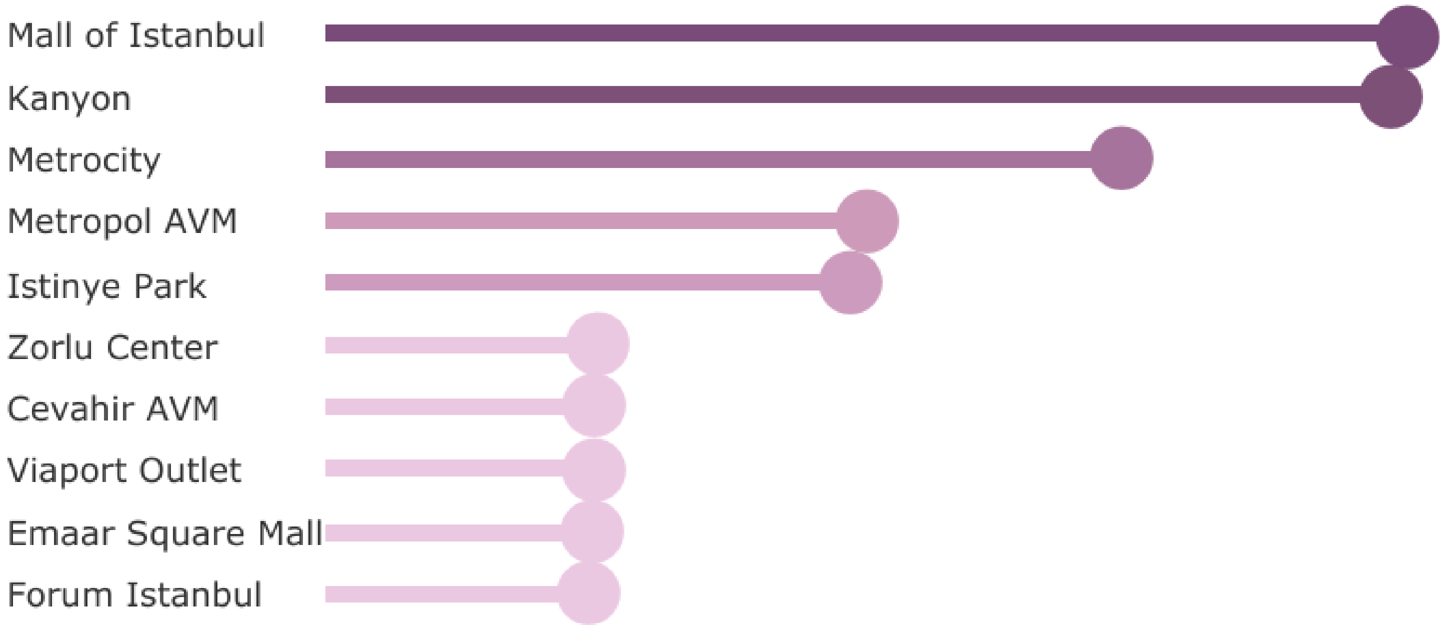


Export As

Total Quantity Sold | Low Revenue Category Focus



Sales by Shoping Mall



Follow

Sales Trends & Revenue Drivers

Exploring category-wise performance to highlight the backbone of total sales

Navigate

Export As

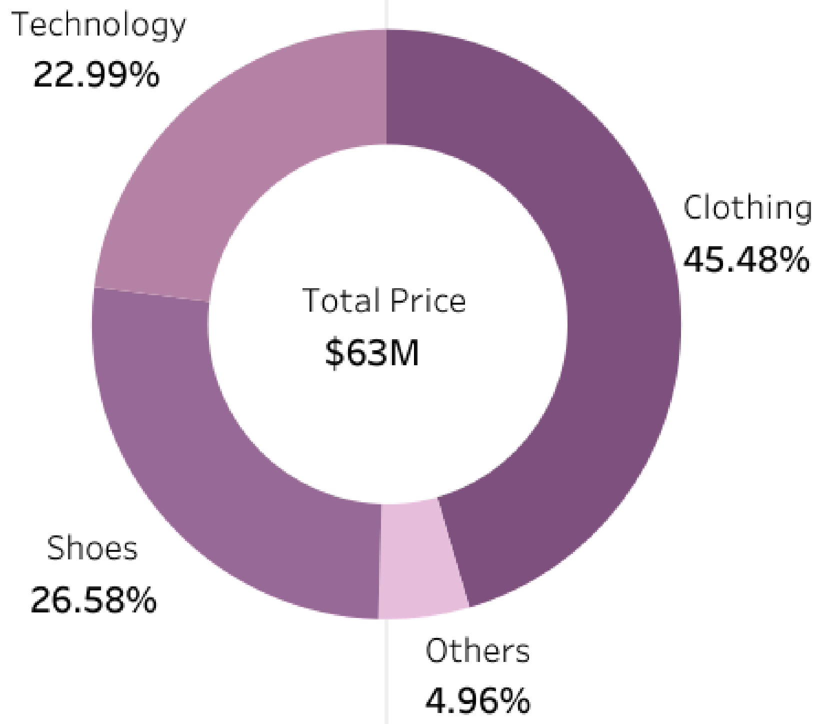
Follow

62,689,131
Contribution of **Low Revenue Categories** on Overall Sales
4.95%

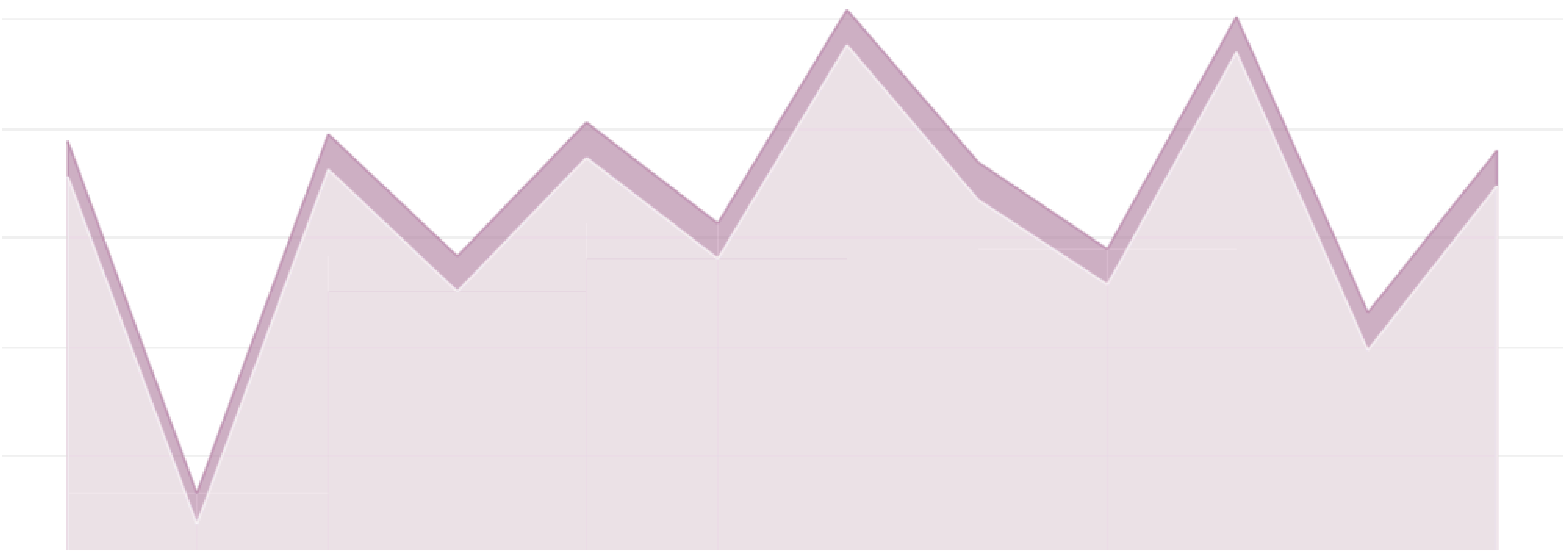
Total Sale Price
$63M

Top 3 **High Revenue Categories** Contribution
95.05%

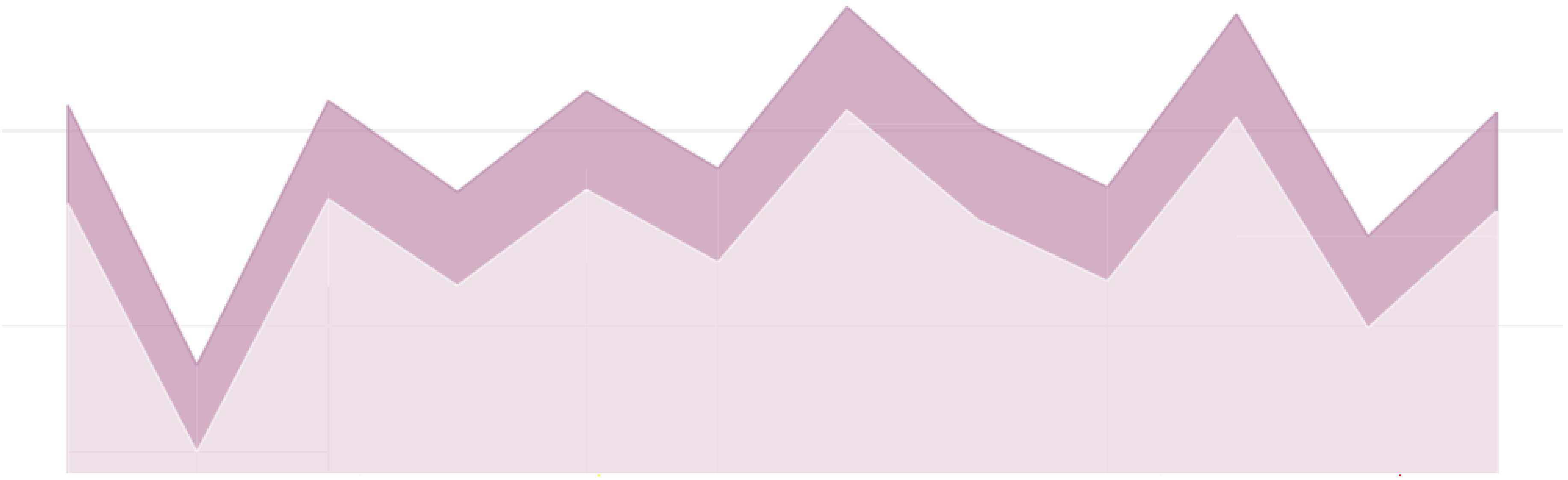
Top 3 Categories | Sales By Category



Impact of Price Simulation on Low Revenue Categories | 20-25% Simulation



Impact of Price Simulation on High Revenue Categories | Only 5% Simulation



Contact me

If you are looking for a detail-oriented data analyst, I would be happy to discuss how I can add value to your team

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